

Federated Learning for Healthcare: Evaluating Client Contribution and Fairness with Heterogeneous Medical Data

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Abstract

Federated Learning (FL) is an emerging machine learning paradigm that enables model training across a large, diverse set of institutions without requiring data centralization. Its main application currently is in the healthcare sector, where concerns about patient confidentiality, ethics, and regulatory compliance, provided by HIPAA and GDPR, prevent sensitive medical histories from being centrally stored. This work focuses on an investigation into FL using a COVID-19 chest X-ray dataset. It considers the performance evaluation of four CNN architectures across a range of heterogeneous data partitions, including label skew, feature skew, concept shift, and Dirichlet-distributed partitions, as well as IID baselines. This work will analyze the impact of non-IID distributions on model performance. It extends fairness evaluation by introducing a definition of client contribution (measured by changes in leave-one-out accuracy) and evaluating client benefit (measured by accuracy gains resulting from collaboration). These experiments show asymmetries in contributions and benefits among clients in GPU-enabled Flower framework environments, highlighting the need for fair aggregation and incentive mechanisms. It also aims to provide an interactive visualization system that facilitates comparisons of relative contributions and rewards across a wide range of architectures and scenarios. Overall, this work provides a systematic empirical assessment of federated learning applied to heterogeneous healthcare datasets, extends the investigation of fairness by deploying dual contribution and benefit metrics, and provides tools and perspectives developed to support clinically appropriate and equitable federated learning.

Contents

Abstract	I
Contents	II
List of Figures	V
List of Tables	VI
List of Abbreviations	VII
1 Introduction	1
1.1 Background and Motivation	1
1.2 Federated Learning in Healthcare	2
1.2.1 History of FL in Healthcare	2
1.2.2 Importance of Privacy-Preserving Approaches	3
1.3 Research Problem and Objectives	4
1.4 Research Questions and Hypotheses	5
2 Literature Review	7
2.1 Overview of Federated Learning	7
2.1.1 Centralized vs. Decentralized Learning	7
2.1.2 FL Architectures and Key Principles	8
2.2 Federated Learning in Healthcare	9
2.2.1 Early Research and Applications	9
2.2.2 Privacy and Ethical Considerations	10
2.3 Fairness and Client Contribution in FL	12
2.3.1 Studies on Contribution and Benefit Metrics	12
2.3.2 Fairness and Incentivization Mechanisms	13
2.4 CNN Architectures for Medical Federated Learning	15
2.4.1 Use of CNNs in FL	15
2.4.2 Comparative Analysis of Existing Approaches	17
2.5 Summary of Research Gaps	19
3 Dataset and Preprocessing	20
3.1 COVID-19 Chest X-ray Data Collection and Sources	20

3.2	Data Preprocessing	21
3.2.1	Cleaning and Augmentation	21
3.2.2	Normalization and Resizing	22
3.3	Dataset Partitioning and Challenges	23
4	Methodology	26
4.1	System Architecture	26
4.2	Federated Learning Workflow	26
4.2.1	Local Training and Weight Aggregation	27
4.3	Convolutional Neural Network Models	28
4.3.1	TinyCNN	29
4.3.2	SimpleCNN	30
4.3.3	ResNet18	31
4.3.4	MobileNetV2	32
4.4	Evaluation Metrics	33
4.4.1	Client Contribution (Leave-One-Out Accuracy Drop)	33
4.4.2	Client Benefit (Accuracy Gain)	34
4.4.3	Global Model Accuracy	35
4.5	Implementation Details and Reproducibility	36
4.5.1	Phase 1: Dataset Acquisition, Preprocessing, and Partitioning into Heterogeneity Scenarios	36
4.5.2	Phase 2: Isolated Training for Each Client across All CNNs and Data Splits	36
4.5.3	Phase 3: Federated Training across All Combinations of Mod- els and Data Scenarios using Flower	37
4.5.4	Phase 4: Logging, Metric Aggregation, and Performance Eval- uation	38
4.5.5	Phase 5: Client Contribution and Benefit Computation	38
4.6	Summary of the Methodology	39
5	Experimental Results and Analysis	40
5.1	Results by Model and Scenarios	40
5.1.1	TinyCNN across Data Scenarios	40
5.1.2	SimpleCNN across Data Scenarios	41
5.1.3	ResNet18 across Data Scenarios	42
5.1.4	MobileNetV2 across Data Scenarios	42

5.2	Client Contribution Analysis	43
5.2.1	Leave-One-Out Accuracy Drop Comparisons	44
5.2.2	Patterns Across Models and Scenarios	46
5.3	Client Benefit Analysis	47
5.3.1	Accuracy Gain Comparisons	48
5.3.2	Patterns Across Models and Scenarios	51
5.4	Observations and Key Findings	53
6	Fair Client Selection and Incentivization	55
6.1	Theoretical Background	55
6.2	Simulation Setup for Incentivization	55
6.3	Experimental Results	57
6.4	Insights for Fair Client Selection	59
6.5	Implications for Healthcare FL Deployments	60
7	Discussion and Future Works	62
7.1	Interpretation of Key Findings	62
7.2	Limitations of Current Work	63
7.3	Future Research Directions	64
7.4	Conclusion and Final Thoughts	65
7.4.1	Summary of Contributions	65
7.4.2	Significance of the Study	65
7.4.3	Final Reflections	66
	Bibliography	67
	Declaration of authorship	71

List of Figures

Figure 1: Comparison of training approaches	2
Figure 2: Comparison of three images	22
Figure 3: Federated Learning Client-Server Architecture.	27
Figure 4: Global accuracy over 10 training rounds for an example scenario (e.g., Feature Skew) comparing all four models.	43
Figure 5: Heatmap of client contribution values for the ResNet18 model across all seven data scenarios	46
Figure 6: Federated learning benefit for each client in the Quantity Skew sce- nario (SimpleCNN model).	51
Figure 7: Average fairness correlation per model with error bars indicating standard deviation across data splits.	57
Figure 8: Fairness stability measured as variance of correlation across splits. .	58

List of Tables

Table 1: Summary statistics of client contribution scores across all data splits for each CNN model.	47
Table 2: Benefit statistics across models and data distribution scenarios. . . .	52

List of Abbreviations

FL	Federated Learning
AI	Artificial Intelligence
FLwr	Flower (Federated Learning framework)
GDPR	General Data Protection Regulation
HIPAA	Health Insurance Portability and Accountability Act
EHR	Electronic Health Record
CXR	Chest X-Ray
AUC	Area Under the Curve
CNN	Convolutional Neural Network
GPU	Graphics Processing Unit
IID	Independent and Identically Distributed
COVID-19	Coronavirus Disease 2019
TinyCNN	Tiny Convolutional Neural Network
SimpleCNN	Simple Convolutional Neural Network
ResNet18	Residual Network (18-layer CNN architecture)
MobileNetV2	Mobile Network V2 (CNN architecture)

1 Introduction

1.1 Background and Motivation

Data and machine learning (ML) are becoming more vital in healthcare, enhancing diagnostics, forecasts, and personalized treatments for patients. There are many valid reasons why people are becoming concerned about privacy and ethics when it comes to sharing patient data and information, as many agencies hold patient data within their own hospitals and organizations. Researchers have observed that while collaborative model training across organizations can significantly improve performance by pooling knowledge, *“the biggest concern with collaborative training in healthcare is data privacy”* [1]. In practice, privacy restrictions impede much of what is technically possible with ML [1]. There is thus a strong motivation to develop privacy-preserving machine learning approaches that enable multi-center data collaboration without exposing sensitive information.

One such approach gaining significant attention is Federated Learning (FL). Federated Learning is a distributed learning paradigm introduced by Google in 2016 [1]. In FL, a centralized coordinator (server) aggregates model updates from multiple participants (clients), each of whom trains on their local data; raw data never leaves the local sites. By design, this addresses the privacy shortcoming of traditional centralized learning, in which a single repository collects all data [1]. **Figure 1** illustrates this difference: in local knowledge, models are trained only on isolated data; in centralized learning, systems pool data from all sites on a central server; in federated learning, models are trained collaboratively through iterative parameter exchanges while data remain distributed [1].

Overall, the research background is based on two trends: (1) the increasing demand for multi-center learning to develop strong medical AI models, and (2) the need for privacy protection of sensitive health data. Federated Learning has emerged as a

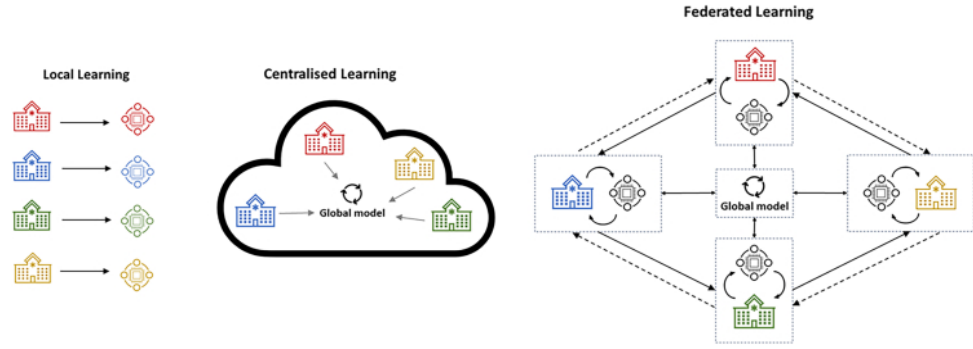


Figure 1: Comparison of training approaches

logical solution to these two trends by supporting collaborative modeling without compromising the privacy of the data used in the modeling process. The following sections will further develop the role of FL in healthcare, providing the background to the specific research questions being addressed in this work.

1.2 Federated Learning in Healthcare

1.2.1 History of FL in Healthcare

Federated Learning (FL) was initially developed for mobile devices, such as training an innovative text model on smartphones without sharing personal data. In a paper published by Google in 2016, the idea of FL was formally introduced and demonstrated that it is possible to train a model in a decentralized manner by transferring model updates rather than sharing data; thus, the name “Federated Learning” was coined [1]. Shortly thereafter, researchers recognized the applicability of FL to healthcare, where data are naturally partitioned across clinics and hospitals. Early proof-of-concept studies around 2018–2020 applied FL to medical imaging and health records, showing that multi-hospital models could achieve accuracy close to that of a centrally trained model while each hospital kept its data locally [2]. In a study by Sheller et al. (2020), one of the earliest reports of multi-institutional federated learning (FL) in medicine was presented, in which researchers trained a tumor segmentation model on MRI scans from different hospitals without sharing patient scans [3]. At about the same time, Rieke et al. (2020) declared federated learning to be the “future of digital health”, addressing opportunities for AI collaboration which had been hindered in the past by concerns for privacy [4].

Since these pioneering efforts, FL has rapidly expanded into various healthcare domains. Notably, medical imaging has emerged as one of the most active research areas for federated learning (FL). To date, studies have employed FL in tasks such as detecting chest X-ray abnormalities, medical image segmentation, and pathology slide classification [1]. Their federated model achieved performance on par with a centrally trained model while each hospital’s data stayed behind its firewall. Beyond imaging, FL has been applied to electronic health records (EHR) data to predict hospitalizations and outcomes [1], to omics data for collaborative genomics analysis, and even to smartphone-collected health metrics in large-scale population studies.

1.2.2 Importance of Privacy-Preserving Approaches

One fundamental challenge of using AI in healthcare is protecting patients’ privacy. Federated learning addresses the foundational challenges of data sharing and confidentiality [5] by creating a system that does not require exchanging raw data across the health institutions. However, federated learning itself cannot prevent data leakage indefinitely or achieve perfect secrecy. To make it more private and reduce potential exposure, it will often require assistance from complementary protection mechanisms. The transmission of model or gradient updates, which is central to the FL process, can inadvertently expose sensitive information through advanced attack methods such as gradient inversion [6].

In healthcare, patient medical information is strictly confidential, and its breach is not only illegal but also creates concerns for the use of artificial intelligence in healthcare in the eyes of the public. Hence, the privacy-oriented approach known as Federated Learning (FL) has emerged in support of the ethical principle of privacy by design [1]. While centralized data breaches are minimized, data is stored on users’ premises in FL, and empirical performance comparisons with traditional centralized models indicate that FL models perform favorably. FL models achieve comparable accuracy while providing significantly greater privacy protection [2]. FL allows patients and research subjects to “*remain in control of their data and still contribute to advancing science*” [2]. This tension has long been challenging to achieve in practice.

1.3 Research Problem and Objectives

Federated learning opens the door to multi-center medical AI, but it also raises unanswered challenges. A core issue, and the focus of this research, is evaluating each client’s contribution and ensuring fairness in federated models trained on heterogeneous medical data. In real-world healthcare federations, clients (e.g., hospitals or clinics) often have non-IID data; their patient populations differ in demographics, disease prevalence, and imaging equipment, among other factors. Recent studies have started to investigate contribution metrics, such as Shapley value-style methods that associate a value to each client based on marginal contributions caused by adding that client’s data [7]. Other heuristics (e.g., Federated Client Contribution Evaluation through Accuracy, FedCCEA) estimate each client’s contribution to global accuracy by re-training or performing gradient analysis [8].

The research problem addressed in this study is: How can we rigorously evaluate client contributions in federated learning for healthcare and utilize this information to improve or ensure fairness in the presence of heterogeneous (non-IID) medical data? This problem sits at the junction of collaborative ML, incentive design, and ethical AI in healthcare. To address this issue, we need to take three essential steps: (1) establish clear and meaningful metrics to evaluate contributions and fairness; (2) analyze how federated training performs with imbalanced or skewed data distributions; and (3) design mechanisms to reduce unfair outcomes. The objectives of this thesis are therefore formulated as follows:

1. **Quantify Client Contributions:** Develop or use a technique to assess the contribution of each client to the overall performance of the federated model. This could involve, for example, using Shapley values, influence function analysis, or measuring the drop in accuracy when a client’s data is removed from the training set.
2. **Assess Fairness Under Heterogeneity:** Establish metrics for fairness in federated learning and apply them to evaluate the trained models. This performance analysis involves performance fairness (accuracy parity across patient subgroups or clients) and contribution fairness (the degree of alignment between a client’s contribution and the benefit received) [9].

3. **Identify Factors Influencing Fairness:** The research investigates the influence of characteristics such as the amount of data, the skewness of the data distribution, and client behavior in terms of participation frequency and hyperparameter selection on client contributions and outcome fairness in general. These correlations make apparent the sources of unfairness in federated learning techniques.
4. **Propose Fairness-Enhancing Strategies:** Based on the insights above, propose and evaluate one or more strategies to improve fairness in federated learning. Possible methods include using weighted federated aggregation, where clients are weighted inversely to their data advantages, fair client selection protocols, and post-training model adjustments to improve performance on the client’s underrepresented patient data.

1.4 Research Questions and Hypotheses

Building on the above problem statement, the research is guided by the following specific research questions and hypotheses:

1. **Contribution Evaluation** – How can we assess each client’s contribution in a federated learning model trained on multi-institutional medical data? The goal is to identify practical strategies to evaluate whether the model’s performance has improved or declined, thereby helping to determine the contribution of each factor in the federated learning model-building process.
2. **Impact of Heterogeneity** – In what ways does data heterogeneity (non-IID distributions across clients) affect the global model’s performance and fairness? This question examines the relationship between differing client data characteristics and outcomes, such as model accuracy per client, convergence speed, and any bias the global model exhibits toward dominant data sources.
3. **Fairness Mechanisms** – What mechanisms can be employed to ensure a fair training process and outcome in federated learning, and how effective are they? This involves investigating methods, such as fair aggregation algorithms and incentive systems, to determine if they can improve fairness scores, i.e.,

reduce performance gaps between high-resource and low-resource clients, while maintaining strong performance across the board.

Hypotheses:

1. **H1:** It is hypothesized that a small subset of clients will account for a disproportionately large contribution to the federated model’s performance. In other words, due to data heterogeneity, some clients will have outsized influence on the model, as measured by contribution metrics. Computing contribution scores will test this hypothesis for each client; it is expected that contributions will be skewed rather than evenly distributed among participants.
2. **H2:** Non-IID data will lead to fairness disparities in the baseline federated model. Specifically, the hypothesis is that under standard FedAvg training, the global model will perform significantly better on data distributions that are well-represented (or dominant) among the clients than on those that are under-represented. For instance, if one hospital’s data constitutes 50% of the training pool, the global model may be biased toward that hospital’s patterns. It is anticipated that observing differences in per-client accuracy or other fairness metrics will indicate this imbalance.
3. **H3:** Incorporating contribution evaluation into the federated process can improve fairness. It is hypothesized that by leveraging knowledge of each client’s contribution, the training process can be adjusted to yield a more balanced model. In practice, strategies such as contribution-based weighting and fair client selection should reduce performance dispersion across clients, thereby improving accuracy parity and overall model quality.

In summary, this introduction has carefully laid the groundwork by emphasizing the significant relevance of federated learning in the healthcare context. It examined the various challenges related to fairness in this new context and elaborated the key questions this research intends to answer. Building on this foundation, the following chapters will expand the discussion to provide a detailed examination of federated learning, focusing on evaluating client contributions and exploring fairness issues in depth.

2 Literature Review

2.1 Overview of Federated Learning

Federated Learning is a decentralized machine learning in which several clients collaborate to learn one model with the assistance of a server at the center [10]. No single server or data center holds the training data in centralized learning; instead, each server or data center holds its own training data. The general concept behind FL is that the data are localized (e.g., on hospitals' or users' devices). However, the model's parameters/updates are shared, thereby preventing privacy leakage and complying with data governance prohibitions [4]. The concept was introduced by McMahan et al. (2016) through the FedAvg algorithm [10], which explores the possibility of learning deep networks in a distributed manner while still achieving robust performance despite non-IID and imbalanced data across clients [10]. FL was initially designed for cross-device scenarios (e.g., training AI models on millions of mobile phones in a privacy-preserving manner) and has since been extended to *cross-silo* settings, such as healthcare, where a modest number of organizations (hospitals, labs) collaborate. Next, the key distinctions between centralized and federated learning are outlined, providing an overview of the architectures and principles of FL.

2.1.1 Centralized vs. Decentralized Learning

In a centralized learning setup, data from multiple sources is aggregated into a single repository or data lake, and a model is trained on this unified dataset. Although this approach consolidates all data, it poses significant challenges to privacy, security, and compliance in the healthcare sector. Relocating sensitive patient data to a single centralized location will inevitably expose patients to legal violations (e.g., GDPR)

and make it easier to compromise their data [11]. On the other hand, federated (decentralized) learning does not require the parties (Clients) to put their data on the internet, as the data remains behind their firewall. At the same time, only model updates (e.g., gradients or weight increments) are sent to a central aggregator [4][10]. With on-device or on-premises training, FL is intentionally designed to support privacy preservation and address pertinent data governance issues associated with centralized learning.

2.1.2 FL Architectures and Key Principles

At its core, FL follows an iterative distributed training process. In each round of training, a subset of clients is selected (e.g., based on availability or to ensure a fraction C of clients per round [12]). The current global model is then sent to each client, which performs local training on its private data for a specified number of epochs. Then, the clients send their model updates, either the trained weights or the computed differences in gradients, to the server, which aggregates them (for example, through simple averaging, as in FedAvg [10]) and updates the global model. The next round begins [12]. This process is repeated iteratively until either the stopping criteria are met (e.g., after a certain number of rounds or upon convergence). At the algorithmic level, FedAvg and most of its variants operate under the assumption of synchronous rounds. However, asynchronous updating architectures have also been analyzed (in which clients send updates intermittently without requiring others to wait) [13]. The client’s selection strategy, synchronization method, and data-aggregation rule are FL’s chief design features [13].

A fundamental challenge arising from FL’s decentralized nature is data heterogeneity: clients typically have non-IID (i.e., not independently and identically distributed) data, meaning each client’s data distribution may differ, sometimes drastically, from others. This can lead to fairness and convergence issues that are absent in centralized training [13]. The analysis examines non-IID data and how articles have addressed the problems it causes (e.g., by modifying the optimization, such as with FedProx and/or data-sharing techniques). For now, it is sufficient to recognize that addressing heterogeneity is a significant topic in FL system design. Indeed, Kairouz et al. (2021) identified that federated learning “*presents fairness-related challenges not present in the centralized training regime*”, while also offering new

opportunities to tackle those challenges through novel coordination and incentive mechanisms [13].

In summary, FL architectures range from the standard server-client model to fully decentralized networks, but all adhere to the “*bringing the code to the data*” paradigm rather than aggregating data in a single location. The following sections will explore how this paradigm has been applied in healthcare and what unique considerations (such as privacy and fairness) arise in that domain.

2.2 Federated Learning in Healthcare

The health sector soon realized the potential of federated learning to enable multi-site collaborations among institutions in AI without compromising patient data confidentiality [4]. Medical data, such as clinical notes, medical images, or genomic sequences, is highly confidential and centralized in some hospitals or regions. Centralized learning in the classical approach involves pooling such data, which is almost always impossible due to privacy regulations and proprietary concerns [4]. FL provides an attractive solution: hospitals can jointly train, for example, a disease diagnosis model by sharing model parameters instead of patient records. This section reviews early research applications of FL in healthcare, highlighting the privacy and ethical considerations raised.

2.2.1 Early Research and Applications

One of the earliest demonstrations of FL in healthcare occurred in medical imaging. Sheller et al. (2018) conducted a landmark feasibility study on federated brain tumor segmentation across multiple hospitals [4]. In that study, each institution trained a 3D CNN (U-Net) on its own MRI scans of glioma tumors, and a central server periodically aggregated the model weights. The federated model surprisingly achieved tumor segmentation accuracy approaching that of a model trained on pooled data, showing that FL on a realistic medical task can achieve performance similar to centralized training [4]. In line with this, Li et al. (2019) examined federated learning for brain MRI segmentation, which preserves privacy. They based their conclusions on the assumption that multi-site-trained models would perform

better than single-site-trained models [4]. Earlier studies in medical imaging have demonstrated that FL is not only technically feasible but also practical for improving model generalization by leveraging diverse data from multiple centers [4].

Beyond imaging, federated approaches have been tested on electronic health records (EHR) and other health datasets. Huang et al. (2019), for instance, employed FL in distributed electronic medical records for mortality prediction, concluding that the prediction of patient outcomes improved among hospitals in FL sharing local data, rather than relying locally on their data [4]. In another significant study, Chang et al. (2018) demonstrated a distributed deep learning network among institutions for analyzing chest X-ray images [14]. While that work predated the broader use of the term “federated learning”, it implemented a functionally similar paradigm (model averaging across hospitals). It provided early evidence that hospitals could jointly train diagnostic models without exchanging data. For example, Florescu et al. (2022) employed federated learning in combination with pre-trained deep CNNs to detect COVID-19 in CT scans. They reported that their federated learning achieved clinical sensitivity similar to that of a centralized model whilst retaining the locality of each clinic’s data [15]. This demonstrates how transfer learning and FL can be combined in healthcare. Each hospital fine-tuned a pre-trained CNN on its data, and a server aggregated the model updates.

In summary, the early applications demonstrated feasibility with real-world benefits, for example, that models trained via FL could demonstrate high accuracy by learning from broader populations than was inherent to data from a single hospital, and at least two of the studies demonstrated that these benefits were achieved with no aggregation of raw data, thus yielding a regulated-compliance posture. The following subsection will discuss these privacy considerations in detail, as well as the ethical concerns associated with a federated approach in healthcare.

2.2.2 Privacy and Ethical Considerations

Privacy is paramount in healthcare ML, and FL is inherently designed to address privacy by keeping data decentralized. By ensuring “the data is not exchanged” between parties, FL *will not risk leaking the user’s privacy or affecting data compliance (such as GDPR)* [15]. This is a compelling advantage over traditional data-sharing. In early deployments of FL within healthcare, this advantage was straightforward.

For instance, in a 2020 multi-center study regarding pathology reports, participating hospitals were willing to participate specifically because the patient data did not leave their possession [4]. FL also adheres to ethical guidelines that require data minimization, sharing only the minimum information necessary. In FL, the only information leaving a site is abstract model parameters or gradients, which are far less interpretable than raw patient data.

However, it is crucial to recognize that FL alone is not a silver bullet for privacy. Researchers have shown that model updates can still leak information about the underlying data through various attacks (e.g., gradient inversion attacks or membership inference). For instance, it has been demonstrated that one can sometimes reconstruct sensitive inputs (like imaging data) from gradients – *“it is possible to reconstruct a patient’s face from CT or MRI data”*, as Rieke et al. note, even if obvious identifiers are removed [4]. This highlights that merely not sharing raw data does not guarantee confidentiality; additional safeguards are often required.

Ownership and control of data are yet another ethical dimension. FL permits institutions to retain control over their data, which should reduce concerns about potential malfeasance involving shared data. Jointly affiliated hospitals can agree to build a model together without any one institution giving up a copy of its dataset. This is an advantage of FL in terms of legal and trust-based concerns. For instance, the Mayo Clinic, along with other health systems, has achieved success with federated learning (FL) in imaging, mainly because each party’s data is on-premises and under that party’s control (as cited in various case studies from 2021 to 2022). This model of collaboration raises additional questions about how to ensure parties behave honestly (i.e., do not poison the model), how to audit contributions, and how to equitably share the benefits of the trained model.

In summary, federated learning offers a promising approach to upholding privacy and complying with strict healthcare data regulations, which is why it has seen early adoption in the field. However, to fully satisfy ethical principles, FL implementations often augment the basic approach with cryptographic protocols, differential privacy, and carefully designed agreements on model usage. The following section will focus on another essential aspect: fairness and client contribution in FL, which guarantees that the collaboration is fair and each participant’s role is appropriately recognized.

2.3 Fairness and Client Contribution in FL

Federated learning presents specific challenges and opportunities regarding fairness, both in the performance of models across participants and in the credit or rewards each participant receives for their contributions. Whereas in centralized learning, there is a single agent who owns all the data and thus reaps all the benefits, FL involves multiple parties with different amounts and types of data and resources. This raises questions, such as: Are some clients disproportionately influencing the model? Do all clients benefit fairly from the global model? How can we quantify each client’s contribution to the federated model, and should rewards (or model access) be tied to these contributions? A growing body of research over the past few years has tackled these questions. In this section, studies on measuring client contributions and benefits (Section 2.3.1) and mechanisms to ensure fairness and proper incentives in FL (Section 2.3.2) are reviewed.

2.3.1 Studies on Contribution and Benefit Metrics

Measuring client contribution: When collaborative learning is utilized, gauging the degree to which each client contributed to the accuracy of the final model is significant. A popular metric is to assess the impact of each client’s dataset on the global model. One straightforward metric is the leave-one-out (LOO) approach, in which one client’s contribution is removed from the training set. The accuracy (or loss) is documented when the client is removed, and the drop in global model accuracy or increase in loss is evaluated on the validation set [12]. A larger drop indicates that the client’s dataset was more valuable when creating the global model. This marginal contribution idea was used by Wang et al. (2019), who proposed a “grouped instance influence” method to measure contributions in horizontal FL [12].

A more rigorous framework for contribution measurement is provided by cooperative game theory, specifically the Shapley value. The Shapley value provides an axiomatic method for distributing credit among participants by averaging their marginal contributions over all possible inclusion orders. Several works have adapted Shapley values to the framework of FL. For example, FL-Shapley (as discussed by Sim et al., 2020, and others) calculates each client’s value by retraining the global model with

and without that client (and combinations thereof). In theory, Shapley values satisfy fairness properties, e.g., two identical clients (with similar data) receive equal value, and any client that never improves the model gets zero credit [16]. Xu et al. (2021) and other researchers have considered Shapley-based metrics in FL, interpreting the federation as a game in which clients “cooperate” to build the model [17].

Measuring Client benefit: Fairness in FL also involves looking at outcomes for each client. One notion of fairness is that each client should benefit from participating, i.e., the federated global model should outperform the client’s locally trained model on that client’s own data. If a client sees no improvement (or worse, a decline) after joining the federation, the collaboration may be deemed unfair or, at the very least, unattractive to that client. This has led researchers to evaluate benefit metrics, such as the accuracy gain: for each client, compare the global model’s accuracy on that client’s test data to the accuracy of a model the client could train in isolation [12]. Nishanbayev et al. (2020) formalized this as “benefit ratio.” In this research, a similar idea is used: each client’s benefit = (federated model accuracy on client’s data) – (client’s standalone model accuracy), as indicated earlier. This captures whether collaboration is mutually beneficial for all parties involved. Ensuring that every client, even those with less data, obtains a positive benefit can be considered a form of collaborative fairness [16].

In summary, numerous metrics have been proposed to quantify client contributions (marginal loss, Shapley value, etc.) and benefits (accuracy gain, etc.). These metrics are instrumental for diagnosing fairness in FL. They are also the foundation for designing incentive schemes, to which we now turn. After all, if a client’s contributions can be measured, one can tie rewards or participation incentives to them, encouraging data holders to join FL and “rewarding those who provide high-quality data and actively engage” [16], while “punishing participants for bad behavior” (e.g., free-riding or contributing low-quality updates) [16].

2.3.2 Fairness and Incentivization Mechanisms

Ensuring fairness in FL often comes down to designing proper incentive mechanisms. Each client in a federation incurs costs (computational, communication, privacy) to participate, so they must have a sufficient incentive, typically, the benefit of a better

model or a tangible reward, to collaborate. A couple of representative approaches include:

- **Reward sharing based on contributions:** A simple idea is to pay or credit clients based on their measured contributions (using metrics in Section 2.3.1). Take, for example, if one were to apply Shapley value or leave-one-out evaluation, then if Client A contributed 20% of the model’s performance improvement, Client A would receive 20% of a reward pool or computing credits. Profit allocation via Shapley has been studied by Song et al. (2019), and others [16]. A challenge here is that contributions can overlap (two clients might have redundant data; each alone is not critical, but together they are important), so any strict ranking must be handled carefully.
- **Reputation and scoring systems:** Kang et al. (2019) introduced a *reputation-based scheme* in which the server maintains a reputation score for each client that updates over time based on the quality of their model updates [12]. Clients that have a stronger reputation (in other words, they regularly submit useful updates) get more opportunities for selection in later rounds or are used to get better rewards. Thus, this principle, in a certain sense, “punishes” bad behavior: suppose the client sends a pointless or corrupt update; in that case, it negatively impacts their reputation and may result in exclusion (to protect the overall integrity of the model).
- **Fair Federated Optimization:** Another approach is to incorporate fairness directly into the model training objective. It is also noted that q-FedAvg and Agnostic FL, as earlier [18], guarantee that the resultant model is equally effective (e.g., by maximizing the lowest client accuracy). There is also research on fair aggregation, in which the server gives greater weight to underrepresented clients during aggregation so their information is not left out. This can prevent unfairness where the global model is biased towards majority data and struggles with data from minority clients [18]. Jiang et al. (2019) and others refer to this endeavor as the field of “fair federated learning,” aiming to achieve balanced error rates across all clients.
- **Collaborative fairness and reciprocal incentives:** Lyu et al. (2020) introduced the concept of “*collaborative fairness*,” which requires that gains be proportional to contributions [16]. Following this work, in the recent research

by Zeng et al. (2022), titled “*You Get What You Give*,” the authors introduce reciprocally fair FL in which the objective is that every client’s gain be commensurate with what it contributed to the federation [17]. They modify the training process to do so, weight the free rider’s updates, and make the model learn better from the others’ contributed data. This method closely implicates the incentive (a better model given the data one owns) to contribute.

In conclusion, several incentive mechanisms have been proposed to promote fairness in federated learning. Some of these include payout mechanisms based on contribution measures, algorithmic calibrations that balance model performance, reputation mechanisms, and game-theoretic mechanisms. This research work is particularly concerned with the evaluation side of this equation: quantifying contributions and benefits (as discussed in 2.3.1) under realistic heterogeneity, which can then inform the design of any needed fairness interventions.

2.4 CNN Architectures for Medical Federated Learning

Medical deep learning commonly relies on Convolutional Neural Networks (CNNs), particularly for image-centric applications (such as X-rays, MRIs, and pathology slides) and specific structured data applications. In healthcare, CNN-based models are the obvious choice due to their established performance in image-based classification, segmentation, and detection. This section discusses how CNN architectures have been used in medical FL (2.4.1) and contrasts current methods, including any architectural or methodological compromises made to accommodate the federated environment (2.4.2). A particular focus is on how different network complexities (from small, custom CNNs to large, pre-trained networks) perform in federated training and on the gaps in comparative evaluations.

2.4.1 Use of CNNs in FL

CNNs have been the workhorse for medical image analysis for over a decade, and this remains true in federated learning projects. Early federated medical studies (as mentioned in Section 2.2) used CNN variants tailored to specific tasks, such as U-Net

and similar encoder-decoder CNNs for 3D MRI segmentation [4], and classification CNNs (e.g., ResNet or VGG) for tasks such as chest X-ray disease classification [4]. In Sheller et al.’s 2018 federated brain tumor segmentation, each site trained a 3D U-Net model locally; the federation process aggregated those CNN weights to build a robust global segmentation model [4]. In another example, *Chang et al. (2018)* used a standard DenseNet architecture, a type of convolutional neural network (CNN), to classify thoracic diseases on chest X-rays. They implemented distributed training across multiple hospitals (comparable to FL) that achieved performance on par with that of centrally trained DenseNets [4].

In many medical FL experiments, researchers have tested multiple CNN models to determine which performs best in a federated setting. For example, Brisimi et al. (2018) (an earlier distributed learning work) tested both a multi-layer perceptron and a CNN for a diagnosis task across hospitals, finding the CNN significantly outperformed the MLP when data included images, even under the federated training regime. In federated COVID-19 diagnosis from CT scans, researchers often compare models such as VGG-16, ResNet-18, and MobileNet (some of which are lightweight CNNs suitable for edge devices). One federated COVID-19 X-ray study (e.g., Liu et al., 2021) reported that MobileNet (a small CNN) trained faster under FL but achieved slightly lower accuracy than ResNet-18 (a larger CNN). Conversely, ResNet-18 achieved higher accuracy but at the expense of more communication rounds. These comparisons reflect the typical accuracy-efficiency trade-off in FL: larger CNNs can better leverage diverse federated data (larger capacity), but may require more rounds or additional bandwidth to achieve convergence.

Overall, CNNs remain at the core of federated learning in healthcare, and to a large extent, the same architectures are used in standalone deep learning applications in FL. The primary variance arises from practical constraints (model size vs. bandwidth) and the need to adjust training hyperparameters (e.g., learning rates) to achieve convergence in the federated environment. The following subsection compares current methods and research, detailing their methodologies and identifying gaps in existing knowledge regarding how model architecture affects data heterogeneity and federated learning.

2.4.2 Comparative Analysis of Existing Approaches

To date, a variety of federated learning studies in healthcare have been conducted, each with different combinations of data distributions, model architectures, and FL algorithms. A comparative analysis reveals a few key dimensions along which these approaches vary:

- Handling of non-IID data:** Approaches differ in how they mitigate the performance degradation caused by heterogeneity. For example, some studies run FedAvg and report the drop in accuracy on non-IID splits compared to IID splits. In contrast, others introduce methods like FedProx (which adds a regularization term to keep local models closer to the global model) or partial weight sharing. In an empirical comparison by Gurtaj et al. (2023), it was *“emphasized (that there is an) obvious discrepancy between the IID and non-IID instances of accuracy”*, and various FL methods (FedAvg, FedProx, personalized FL, etc.) were benchmarked under different data skew conditions [15]. They found that methods explicitly designed for heterogeneity (such as personalized FL algorithms) yielded more consistent results across clients than vanilla FedAvg, albeit at the cost of some added complexity. However, many medical FL papers have not extensively explored heterogeneity scenarios – typically, each focuses on a single scenario (either IID or a specific type of skew). This indicates a gap: a systematic evaluation across many heterogeneity types (label skew, quantity skew, etc.) has been lacking, something our thesis aims to address.
- Model architecture and performance:** Few works directly compare different CNN architectures in the same FL setting. An exception is a study by Brisimi et al., who evaluated both linear models and CNNs for federated cardiovascular risk prediction (finding that CNNs performed better with enough data). Another example is the federated analysis of COVID images, which utilized both ResNet and MobileNet. The 2021 Engage AI challenge in federated tumor detection saw some participants using simple models to minimize communication, while others employed more complex models and fewer rounds of communication. The results did not yield clear-cut findings, but instead suggested an optimal balance between the two approaches. In general, it is observed that model complexity can exacerbate data heterogeneity: a highly

complex model may overfit the specificities of each client’s data during local training, potentially leading to a divergent global model when the data are non-IID. This research compares TinyCNN, SimpleCNN, ResNet18, and MobileNetV2 under the same federated conditions, a step toward addressing this gap.

- **Federated vs. Centralized vs. Single-Site performance:** Many papers report that federated models achieve performance close to centralized training (where data is pooled). For example, Sheller et al. reported only a slight disparity in the Dice score between their federated model and the centrally trained model [4]. Likewise, Rieke et al. mention papers in which FL achieves AUCs within some small epsilon of the centralized AUC [4]. Conversely, Nguyen et al. (2022) in an ocular imaging FL study observed that although FL improved each clinic’s performance with respect to its local model, the disparity relative to a hypothetical centralized model persisted due to differences in distribution. A comparison across studies indicates that FL yields greater relative improvements for smaller partners (i.e., those with lower data-bearing capacity). In the event the latter partner domineers considerably, the centralized model’s superiority derives almost entirely from the latter’s data. Thus, how fair the performance is distributed can depend on partner sizes.

In light of the above, a notable research gap is the lack of holistic comparative studies; most papers introduce a single innovation (e.g., a new aggregation method or an incentive mechanism) and test it in a single setting. There is a need for comprehensive benchmarks that vary data heterogeneity, model type, and fairness/incentive schemes to evaluate what combination works best for a given scenario. In particular, for medical FL, it remains unclear how to choose the “optimal” CNN architecture for a given federation, should one prefer a smaller model that’s easier to federate or a larger one that could ultimately yield higher accuracy? This research contributes to this question by evaluating multiple CNNs (TinyCNN, SimpleCNN, ResNet18, and MobileNetV2 as per the thesis objectives) across a spectrum of data distributions, and observing their relative performance and training dynamics.

2.5 Summary of Research Gaps

In reviewing the literature on federated learning for healthcare, with a focus on client contributions, fairness, heterogeneous data, and CNN models, several research gaps emerge. In short, the current position is that federated learning, as promising as it is for privacy-preserving collaborative modeling in healthcare, still has work to do to ensure fairness and to study the effects of model design in practice. There is room for more nuanced investigations, particularly studies that not only develop new methods but also critically assess when and why imbalances occur in federated training. These gaps form the foundation for the current research.

By addressing these issues, the research aims to inform the development of more equitable and effective federated learning systems for healthcare, where all clients (large or small) are reasonably accounted for, and the selected models are well-suited to the challenges posed by medical data. This, ultimately, will help in designing FL deployments that are both performance-efficient and fair, thereby incentivizing broad participation, a key requirement for successful real-world adoption.

3 Dataset and Preprocessing

3.1 COVID-19 Chest X-ray Data Collection and Sources

The COVID-19 dataset for diagnosis from chest X-ray images is publicly available and can be downloaded from the University of Montreal. Cohen et al. (2020) made the dataset publicly available as a set of chest X-ray (CXR) images for normal and pneumonia classes, with a focus on COVID-19 [19]. The images were sourced from public repositories, including research articles and online image repositories, and no patient-identifying information was included [19]. As of early releases, the COVID-19 Image Data Collection contained on the order of a few hundred images (e.g., an initial report noted 123 frontal CXR images of COVID-19 pneumonia) [19], and it has been continually expanded. In this research, approximately 137 clean COVID-19-positive CXR images are being used. These were high-quality images that did not contain duplicates or poorly scanned images and were taken from the dataset in Cohen et al. This dataset [19] serves as a standard data benchmark when reviewing COVID-19 chest X-rays.

The classification task involved three categories: COVID-19, Viral Pneumonia (non-COVID), and Normal, requiring additional images for the non-COVID classes. To achieve this, chest X-rays of normal lungs and other pneumonia cases were incorporated from a Kaggle Chest X-Ray Images (Pneumonia) dataset [20]. From this dataset, a subset of healthy (normal), covid, and viral pneumonia images was selected as comparison classes for COVID-19 [21]. The viral pneumonia cases represent infections caused by other viruses. By combining these sources, the final dataset consisted of approximately 317 images: 137 COVID-19 cases and around 180 images from different categories (normal lungs and viral pneumonia). Images were divided into training and test sets using stratified sampling by class, and the resulting sets

were split into individual folders. Subfolders for each of the three classes (COVID, Viral Pneumonia, and Normal) were created within these folders to facilitate loading the images and labels during model creation.

3.2 Data Preprocessing

3.2.1 Cleaning and Augmentation

Data cleaning was minimal since the images contained no artifacts or corruption, making them suitable for analysis. No additional filtering or denoising was required, beyond those already included in the dataset. All images were checked to ensure they were correctly labeled and in a view (frontal chest X-ray) suitable for the task; no images were discarded, so no specific cleaning technique was applied.

The primary preprocessing effort involved data augmentation to address the limited number of training samples. Augmentation artificially increases the dataset size by generating multiple transformed versions of each image, helping overcome overfitting and improving generalization to new data [22]. In fact, some recent studies on COVID-19 CXR use augmentation (e.g., random rotations, flips, and translations) to improve the predictive performance of their models [22]. Following this strategy, we implemented an augmentation pipeline that generated five new images per original image (an augmentation factor of $5\times$). Each original CXR in the training set produces five randomized variants using a combination of geometric and intensity transformations:

- **Geometric transformations:** Images were randomly rotated (90° steps or small $\pm 15^\circ$ angles), flipped, and lightly zoomed or shifted (up to 10%) to simulate variations in patient positioning. These augmentations help the model generalize to natural geometric differences in medical imaging.
- **Intensity and noise transformations:** Brightness and contrast were adjusted by $\pm 20\%$, and Gaussian noise or elastic deformation was applied. These simulate exposure differences, sensor noise, and mild anatomical distortions.
- **Occlusion (Coarse Dropout):** Small rectangular regions (5–15% of the image, 1–8 patches) were randomly removed and filled with zeros. This trains

the model to stay robust even when parts of an X-ray are occluded or contain artifacts.

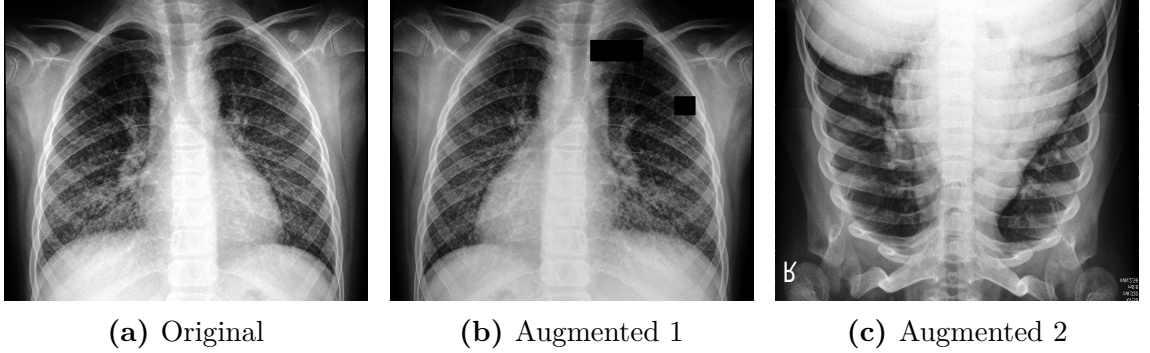


Figure 2: Comparison of three images

Overall, the augmentation pipeline introduced a wide variety of plausible changes while preserving the underlying class label. **Figure 2** illustrates examples of an original CXR and a couple of augmented versions. By generating multiple augmented images for each original, we expanded the training set from 317 to $\approx 1,800$ images (originals plus augmentations). It is essential to specify that augmentation was limited exclusively to the training data; the test set data was not augmented (but was normalized; this will be discussed below), therefore providing evaluation on image data that was actually real (unaltered) images. The purpose of the augmentation strategy was to increase the model’s ability to generalize to new, previously unseen chest X-rays by sampling the model under various transformations during training.

3.2.2 Normalization and Resizing

In addition to augmentation, standard normalization and resizing were performed on the images. Since deep learning models require uniform input sizes, each X-ray image is resized to 224×224 pixels (width $\times$ height). This is a standard input size for well-known CNN architectures (e.g., ResNet) that attempt to balance detail size and computational efficiency [23]. The original chest X-rays had a variable resolution (commonly 1024×1024 or larger). These images were then down-sampled to 224×224 using bi-linear interpolation, which slightly distorts the aspect ratio if the original was not square; however, in our case, most CXRs are roughly square or were center-cropped by the dataset curators, so distortion was minimal. The

resizing ensures that all images can be batched together and that the model has a consistent scale to learn from. Overall, the CXRs are treated as 8-bit images with values ranging from 0 to 255 per pixel, across three channels.

The dataset was prepared for training using a data generator that read images from the directory structure, assigned class labels, and produced batches for model input. The three classes were numerically encoded: Normal = 0, Viral Pneumonia = 1, and COVID-19 = 2, and internally managed via one-hot (categorical) encoding. The generator’s shuffle option was enabled with a fixed random seed to avoid order bias, so that all images were presented in a randomized order across epochs. The preprocessing pipeline was applied identically to all training images, including resizing, normalization, and augmentation. For test images, resizing and scaling were applied to a 224×224 pixel resolution in the $[0, 1]$ range, without any random augmentations, to ensure equal conditions for evaluation.

3.3 Dataset Partitioning and Challenges

After preprocessing, the next step was to split the dataset for federated learning experiments. In FL, data partitioning simulates distributing data to different clients. A key aspect is how the data is separated, especially since real-world data across clients is usually heterogeneous (non-IID). Seven partitioning scenarios are designed to explore different types of data heterogeneity:

- **IID Equal:** In this baseline scenario, the data is split independently and identically distributed among clients. Each of the three clients is given an equal partition of the data, and each partition follows the same class distribution as the overall dataset. In practice, this meant that each client was given a mix of COVID, pneumonia, and a normal X-ray in roughly equal proportions. This is an ideal scenario for FL (no heterogeneity) and is used here as a control.
- **Quantity Skew:** In this situation, every client has data for all three classes, but the number of samples per client varies; for instance, a client may have a large portion of the data, while other clients have very few images. This can be thought of as a situation where a hospital has many patients compared to a small clinic. It is ensured that at least a few samples of each class were

provided per client, but the total quantities were unequal (hence the term “skewed”).

- **Label Skew:** An example of a case where the label class distribution is unbalanced [24]. For each client, the data within a particular class varies in terms of ratio and does not replicate the total data distribution. For example, Client A has a case load primarily consisting of COVID-19 cases, with a small number of non-COVID-19 cases. Client B has a case load mainly consisting of non-COVID-19 instances, with a small number of pneumonia cases, among others. Each client sees all three classes, although in an imbalanced distribution.
- **Dirichlet Label:** This scenario implements label skew using a Dirichlet-based sampling procedure. Class proportions for each client are drawn from a $\text{Dirichlet}(\alpha)$ distribution, where the parameter α controls the degree of heterogeneity across clients. This approach is widely used in FL research to generate controllable non-IID partitions [25]. As with label skew, each client receives a different class distribution; for small values of α , a client typically obtains a high concentration of one class and very few samples from others. Dirichlet-based partitioning introduces randomness into the severity of imbalance and provides a more natural simulation of heterogeneous data than manually assigning class splits. In the experiments, it is treated as a distinct scenario to analyze how a moderately extreme, yet randomly induced, label imbalance influences the training dynamics.
- **Pathological (Extreme Non-IID):** The “pathological” split is an extreme label skew case often cited in literature [26]. In this scenario, each client is primarily specialized in one or two classes and may have none of the others. For instance, Client 1 might have only COVID-19 X-rays, Client 2 only viral pneumonia, and Client 3 only normal—or some similarly extreme partition. This setup, sometimes referred to in the literature as ‘pathological non-identifiable’ (non-IID), can be particularly complicated. No client has a representative dataset for all classes [26]. This scenario tests FL’s robustness under highly non-IID data.
- **Feature Skew:** Feature skew (also known as covariate shift) means that the distribution of input features X varies across clients [24]. Here, each client may have a different data collection environment, which can affect the images. For

example, one client’s images could all come from a single hospital’s machine (with a specific image intensity profile), while another client’s images come from a different machine or population. Concretely, we could imagine that one client has all the X-rays from Dataset-A and another has all the X-rays from Dataset-B, introducing a domain shift. This is partially achieved because the COVID images and the normal/pneumonia images originally come from different sources, split by source, which creates differences in feature distribution. The feature skew scenario challenges the model to handle domain-specific features; a global model may perform sub-optimally if it does not see all feature variations during training.

- **Concept Shift:** Differences in the relationship between features and labels across clients [27]. This can be interpreted as each client having a somewhat different understanding of the task; the concept-shift scenario in our particular setting involved applying different systematic transformations to each client’s images. For example, all of Client 1’s images could potentially be high-contrast (simulating a hospital that utilizes a higher dose of X-ray), all of Client 2’s images would potentially be rotated or flipped slightly (simulating a different orientation of patients), and so on. Here, the images are essentially shifted to the “concept” of each client’s image, while keeping the labels the same. By analogy, the concept shift scenario forces the model to reconcile different representational patterns of COVID or pneumonia across clients. This tests whether FL can learn a global model that covers all local “concepts.”

Overall, the dataset was divided into seven different configurations to specifically examine the effect of data heterogeneity on the classification of chest X-ray images for COVID-19 diagnosis under federated learning. Each configuration introduced diverse heterogeneity in data size, label proportions, or feature and concept expression. Such a structure allowed the assessment of the FL framework in both optimal and complex conditions. In addition, this structure informed the model’s robustness and aspects that may pose challenges in real-world applications involving diverse and non-IID data distributions. Each scenario was analyzed to assess model adaptability and identify strategies to improve performance under heterogeneous data conditions.

4 Methodology

This chapter outlines methods for assessing federated learning (FL) in a health-care setting (Chest X-ray data), with a focus on client engagement and fairness when working with heterogeneous data. It presents the system architecture and the experimental design that underpinned the study.

4.1 System Architecture

Figure 3 illustrates the architecture: a central FL server communicates with three client processes (representing hospitals or clinics) to exchange model parameters. At the same time, raw data remains local on each client. Each client node holds its own subset of the COVID-19 X-ray dataset and trains a local model on that data. The server node begins by initializing the global model, then orchestrates the federated learning rounds as follows: it sends the current global model to all clients, collects local training updates from each client, and aggregates them to produce a new global model. This arrangement also conforms to the standard FL paradigm, in which the server coordinates the overall process while clients train on their own local data. The architecture also ensures data privacy: only the exchanged model weight updates are sent, not the underlying X-ray images or labels, which is crucial for meeting the privacy and confidentiality requirements of healthcare use cases.

4.2 Federated Learning Workflow

In this section, the workflow of the federated learning process is described. To simulate multiple institutions, the Flower framework’s [28] simulation capability was used to define various client instances as separate processes or threads on a single

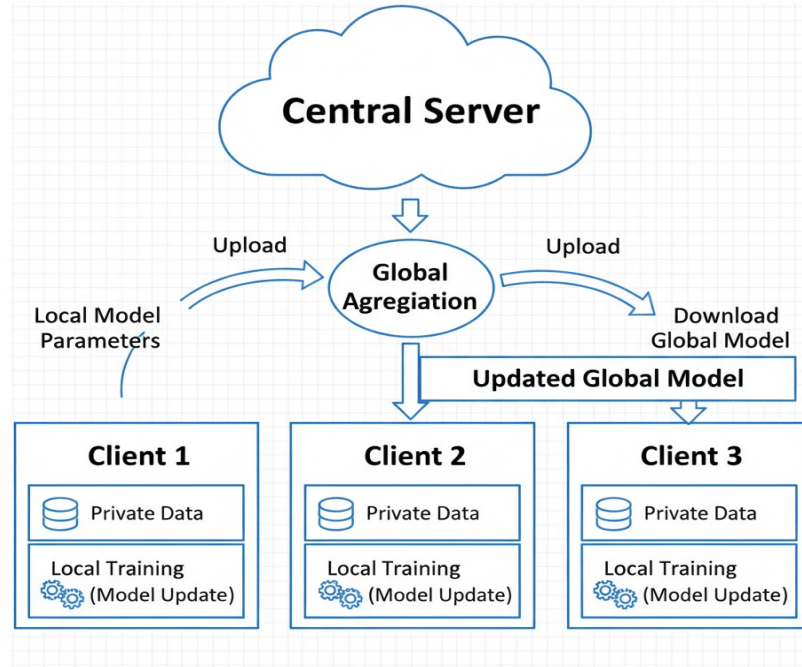


Figure 3: Federated Learning Client-Server Architecture.

system. The workflow follows the classical FedAvg procedure with modifications specific to the experiment (like one local epoch per round).

4.2.1 Local Training and Weight Aggregation

Each client trains on its assigned local data subset and occasionally sends its model updates to a central server that aggregates and averages them. The Federated Averaging (FedAvg) algorithm [29], the most popular training algorithm, serves as the baseline for federated learning. The key steps performed by the client and server are summarized as follows:

- **Client-side local training:** At the start of a round, the server sends the current global model parameters to each client. The client loads these weights into its local model (ensuring all clients start the round from the same model state). Each client then trains the model on its local dataset for a specified number of epochs (in our configuration, one epoch per round for federated training). A single epoch is defined as the point at which the client has iterated

once over all of its local training samples, running forward and backward passes before updating the model weights.

- **Weight update:** After the local epoch, the client updates its model weights to be optimized for its local data. The client then computes a local evaluation (validation accuracy) on its own validation set or a portion of training data, primarily for logging purposes. The client prepares a message containing its updated weights and, optionally, the number of samples it trained on.
- **Server-side aggregation:** The server waits to receive updates from all three clients (synchronous FL). Once all updates are received, the server aggregates the weighted models. Specifically, if w_i represents the weight vector from client i and n_i the number of samples at client i , the server computes the new global weights w_{new} as:

$$w_{\text{new}} = \frac{n_1 w_1 + n_2 w_2 + n_3 w_3}{n_1 + n_2 + n_3},$$

which is a weighted average by data size. This ensures that clients with larger datasets have proportionally more influence on the global update, aligning with the FedAvg algorithm [29].

- **Global model update:** The aggregated weights become the updated global model for the next round. The server may also evaluate this global model on a global validation set or via a weighted combination of client validation metrics to track performance.

Overall, the federated learning workflow employs a standard synchronous FedAvg configuration, with a single local epoch per round and 10 total training rounds. This method strikes a balance between local computation and global synchronization, providing a standard paradigm for evaluating system performance across various data distribution scenarios. The following section provides a detailed explanation of the neural network models used for training in all scenarios.

4.3 Convolutional Neural Network Models

Multiple Convolutional Neural Network (CNN) architectures were evaluated in the federated learning experiment. These included two lightweight custom models developed from scratch (TinyCNN and SimpleCNN) and two established deep architec-

tures with pre-trained ImageNet weights (ResNet-18 and MobileNetV2). All models were configured as three-class classifiers, with an output layer containing three neurons corresponding to COVID-19, Normal, and Pneumonia classes. The architecture and specific training considerations associated with each model are outlined in the following sections.

4.3.1 TinyCNN

TinyCNN is a compact convolutional network designed as a baseline with limited capacity. TinyCNN is intended to have a minimal number of layers, allowing it to train quickly and be less prone to overfitting, given the limited data available to each client. The architecture of TinyCNN is as follows (listing the layers in order):

- **Convolutional block:** 8 filters (3×3 conv, ReLU), followed by a 2×2 Max-Pool.
- After this block, the feature map size is reduced once by max-pooling ($128 \times 128 \rightarrow 64 \times 64$), and the number of channels increases from 3 to 8.
- **Flattening layer:** converts the $8 \times 64 \times 64$ feature map into a 1-D vector.
- **Fully Connected layer (Output):** Linear layer mapping the flattened features to 3 output classes.

```

1 class TinyCNN(nn.Module):
2     def __init__(self, num_classes=3):
3         super(TinyCNN, self).__init__()
4         self.features = nn.Sequential(
5             nn.Conv2d(3, 8, 3, padding=1),
6             nn.ReLU(),
7             nn.MaxPool2d(2), # [8, 64, 64]
8         )
9         # Auto-detect flattened size
10        dummy_input = torch.zeros(1, 3, 128, 128)
11        flat_size = self.features(dummy_input).view(1, -1).size(1)
12
13        self.classifier = nn.Sequential(
14            nn.Flatten(),
15            nn.Linear(flat_size, num_classes)
16        )
17
18    def forward(self, x):
19        x = self.features(x)

```

```
20         return self.classifier(x)
```

Listing 1: PyTorch implementation of TinyCNN

TinyCNN’s performance helps answer whether model complexity is needed to benefit from FL. If TinyCNN performs similarly across scenarios, it might indicate that it is too simple to capture differences. Indeed, it is found that TinyCNN’s accuracy was relatively low across all cases, but it still showed trends in contribution/benefit without the complex interactions of deep networks.

4.3.2 SimpleCNN

SimpleCNN is a slightly more complex CNN built on top of TinyCNN. It has more layers and parameters to improve accuracy, but is still relatively short compared to ResNet-18. The SimpleCNN architecture can be described as:

- **Convolutional block 1:** 32 filters (3×3 conv, BatchNorm, ReLU), followed by a 2×2 MaxPool.
- **Convolutional block 2:** 64 filters (3×3 conv, BatchNorm, ReLU), followed by a 2×2 MaxPool.
- **Convolutional block 3:** 128 filters (3×3 conv, BatchNorm, ReLU), without a MaxPool layer.
- After these blocks, an **AdaptiveAvgPool2d(1×1)** reduces the feature map to size $128 \times 1 \times 1$, regardless of input resolution.
- **Fully Connected layer 1:** Linear($\text{flat_size} \rightarrow 128$), ReLU.
- **Dropout layer** ($p=0.4$): applied before the first fully connected layer.
- **Fully Connected layer 2 (Output):** Linear($128 \rightarrow 64$) + Linear($64 \rightarrow 3$ classes).

```
1 class SimpleCNN(nn.Module):
2     def __init__(self, num_classes=3):
3         super(SimpleCNN, self).__init__()
4         self.features = nn.Sequential(
5             nn.Conv2d(3, 32, kernel_size=3, padding=1),
6             nn.BatchNorm2d(32), nn.ReLU(), nn.MaxPool2d(2), # -> [32, 64, 64]
7             nn.Conv2d(32, 64, kernel_size=3, padding=1),
```

```

8         nn.BatchNorm2d(64), nn.ReLU(), nn.MaxPool2d(2), # -> [64, 32, 32]
9         nn.Conv2d(64, 128, kernel_size=3, padding=1),
10        nn.BatchNorm2d(128), nn.ReLU(),
11        nn.AdaptiveAvgPool2d((1, 1)) # -> [128, 1, 1]
12    )
13
14    dummy_input = torch.zeros(1, 3, 128, 128)
15    flat_size = self.features(dummy_input).view(1, -1).size(1)
16
17    self.classifier = nn.Sequential(
18        nn.Flatten(),
19        nn.Dropout(0.4),
20        nn.Linear(flat_size, 128), nn.ReLU(),
21        nn.Linear(128, 64), nn.ReLU(),
22        nn.Linear(64, num_classes)
23    )
24
25    def forward(self, x):
26        x = self.features(x)
27        return self.classifier(x)

```

Listing 2: PyTorch implementation of SimpleCNN

The reason for including custom models like TinyCNN and SimpleCNN is to analyze client contributions in a setting where model complexity is not a confounding factor from pretraining. These models learn solely from the federated data, so any performance differences across scenarios are attributable to data distribution and training dynamics, rather than external knowledge.

4.3.3 ResNet18

ResNet18 is a well-known deep CNN architecture introduced by He et al. (2016) [30], consisting of 18 layers with skip connections (residual blocks). ResNet18 is primarily used in a pretrained setting, initialized with weights trained on ImageNet. The motivation for using ResNet-18 was to investigate how a high-capacity, pre-trained model performs in federated learning on medical images.

Key characteristics of ResNet18 in our setup:

- **Architecture:** 5 convolutional stages, each with multiple residual blocks. It has ≈ 11.7 million parameters. The network includes Batch Normalization layers, which can be sensitive to differences in data distribution across clients.

- **Pretraining:** Since ImageNet images were very different from chest X-rays, ResNet18 was fine-tuned rather than applied pre-trained. All weights were left trainable, and the model was initialized with pre-trained ImageNet parameters. This provided a strong basis for general feature extraction capabilities, including edge and texture detection, which was especially helpful due to the availability of the average dataset for training.
- **Training in FL:** The same federated learning hyperparameters were applied across all models, with one local epoch per round and a learning rate of $1e-3$. Since ResNet18 is a deeper architecture, each local training epoch required slightly more computation. Training sequentially across three clients remained computationally efficient and well within reasonable compute capacity.

ResNet18 served as a benchmark model in the experiments, providing a reference point for assessing how effectively a robust, high-capacity network can utilize federated data. One challenge was that, with BatchNorm, if a client’s batch contains only one class (pathological split) or has small batch sizes, the statistics can become unstable. In practice, it did not diverge, but it is a known consideration. If it were a bigger issue, one could use GroupNorm or another normalization that does not depend on batch stats [31]. For simplicity, ResNet’s BN is left as it is.

4.3.4 MobileNetV2

MobileNetV2 is another CNN architecture (Sandler et al., 2018) optimized for mobile devices, with depthwise separable convolutions and an inverted residual design. MobileNetV2 is also used with pretrained ImageNet weights. MobileNetV2 is particularly well-suited for federated learning due to its smaller size, enabling more cost-effective communication and local computation. It is included to see if a smaller, pre-trained model could match the performance of ResNet18 and to examine whether the model architecture influences fairness or contribution.

Essential details for MobileNetV2:

- The final classification layer is replaced to output three classes. The rest of the network (feature extractor with convolutional “bottleneck” layers) was kept and fine-tuned.

- Its design includes BatchNorm in each bottleneck, so similar considerations to those of ResNet regarding statistical differences apply.
- The same training strategy is used as ResNet18 (same optimizer, LR, epochs).

One reason to include MobileNetV2 is to evaluate model generalization across clients: sometimes smaller models generalize better with limited data, whereas larger models can overfit to a single client. Federated training with MobileNet might thus be more stable in heterogeneous conditions. In practice, both ResNet18 and MobileNetV2 performed well. The choice between them could come down to efficiency vs. slightly higher accuracy.

In conclusion, four convolutional neural network (CNN) models were trained in the experiments: TinyCNN and SimpleCNN were trained from scratch, while ResNet18 and MobileNetV2 were fine-tuned using pre-trained weights from ImageNet. Having different levels of model complexity enables client contributions and fairness analysis, thereby making the model agnostic. Section 4.4 describes how client benefits and contributions were calculated, and performance was evaluated based on these architectures.

4.4 Evaluation Metrics

4.4.1 Client Contribution (Leave-One-Out Accuracy Drop)

Client Contribution is the drop in global model accuracy when a given client is excluded from training, relative to training that includes all clients. This follows a leave-one-out (LOO) [32] evaluation. Let

$$A_{\text{all}}$$

be the global accuracy using all clients, and

$$A_{(\text{minus-}i)}$$

be the global accuracy when client i is removed. The contribution of client i is:

$$\Delta_i = A_{\text{all}} - A_{(\text{minus-}i)}.$$

A larger Δ_i indicates that the client provides essential data to the global model, while $\Delta_i \approx 0$ implies that the client’s data adds little new or diverse information.

Measurement procedure:

For each scenario, the model is first trained with all three clients to obtain A_{all} . Then, three additional federated runs are performed, each time leaving one client out (e.g., train with clients B and C only to compute $A_{(\text{minus-A})}$). All runs are repeated to account for stochasticity. Global accuracy is calculated on a standard held-out validation set using a weighted combination of per-client results.

This provides an estimate of each client’s marginal performance impact, analogous to simplified Shapley-value reasoning but computed directly via LOO accuracy drops.

4.4.2 Client Benefit (Accuracy Gain)

Client Benefit is defined as the improvement in a client’s model accuracy gained by participating in federated learning, compared to training in isolation on its own data. For client i , benefit is computed as:

$$B_i = A_i^{\text{federated}} - A_i^{\text{isolated}},$$

where $A_i^{\text{federated}}$ is the accuracy of the global FL model evaluated on client i ’s local test set, and A_i^{isolated} is the accuracy of a model trained only on client i ’s data (no federation).

A positive B_i indicates that the client performs better through collaboration, while $B_i \leq 0$ suggests limited or negative benefit.

Measurement procedure:

To compute $A_i^{\text{federated}}$, the best-performing global model (based on validation accuracy) is evaluated on each client’s local test set. To compute A_i^{isolated} , a model with the same architecture is trained solely on client i ’s data for the same number of epochs, and its best accuracy is recorded. The difference between these two values yields B_i .

Interpretation:

High B_i values typically arise when a client has limited, biased, or unbalanced data, since federated learning supplements these gaps with information from other clients. Low or negative B_i may occur if the global model shifts toward a distribution that disadvantages the client. Benefit thus serves as an indicator of fairness, reflecting how much utility a client gains from federation.

4.4.3 Global Model Accuracy

Global model accuracy is used as the primary performance metric. It represents the accuracy of the federated global model when evaluated on a combined validation distribution. Since a single unified validation set is unavailable, global accuracy is approximated by aggregating client-specific validation results using weights. Formally, the weighted accuracy is:

$$A_{\text{global}} = \frac{\sum_{i \in \{A, B, C\}} n_i^{(\text{val})} A_i^{(\text{val})}}{\sum_{i \in \{A, B, C\}} n_i^{(\text{val})}},$$

where $A_i^{(\text{val})}$ is the accuracy of the global model evaluated on client i 's validation data, and $n_i^{(\text{val})}$ is the corresponding number of validation samples.

This formulation yields the accuracy that would be obtained if all validation data were centrally pooled [33]. Global accuracy supports comparisons across distribution scenarios and model architectures and allows tracking convergence across rounds. The final reported value uses the best global model checkpoint (highest validation accuracy).

All metrics are averaged over three independent runs per scenario to reduce variance, and all results are logged for reproducibility. Together with client contribution and client benefit, global accuracy provides a comprehensive view of performance and fairness in federated learning.

4.5 Implementation Details and Reproducibility

4.5.1 Phase 1: Dataset Acquisition, Preprocessing, and Partitioning into Heterogeneity Scenarios

The experimental pipeline began with obtaining a dataset of COVID-19 chest X-ray images and preparing it for federated learning. All images were standardized through preprocessing steps. To enhance model generalization, a moderate amount of data augmentation was applied to the training images. After preprocessing (as explained in Chapter 3), the dataset was split across three simulated client nodes to model federated hospital networks. A fixed random seed was used during this partitioning to guarantee reproducibility of the split. Seven distinct data distribution scenarios were defined to inject controlled heterogeneity across clients. These scenarios included label skew, quantity skew, and other, even more realistic, non-IID scenarios that demonstrate the challenges of real-world data heterogeneity in practice. The research ensured that each heterogeneity condition could be replicated exactly in subsequent runs.

4.5.2 Phase 2: Isolated Training for Each Client across All CNNs and Data Splits

Before federated training, a series of isolated training sessions was conducted for benchmarking. In each session, a single client’s dataset (under a given scenario) was used to train a CNN model from scratch, as if it were a traditional centralized training on that local data. Four CNN architectures of varying complexity were evaluated, ranging from custom, lightweight models (denoted TinyCNN and SimpleCNN) to more complex standard architectures (ResNet-18 and MobileNetV2). Using PyTorch, each model was trained on a client’s training subset independently, employing identical hyperparameter settings across all clients and models to ensure fairness. Specifically, all models utilized the Adam optimizer with an initial learning rate on the order of 10^{-4} (0.0001 to 0.001) and a moderate batch size (32 or 64 images per batch), in line with standard practices for medical image classification. Each local training run lasted long enough to reach convergence on that client’s

data (while avoiding overfitting, given the limited data per client). During training, standard classification loss (binary or categorical cross-entropy, as appropriate to the task) was minimized. The training process for each client-model pair was logged, and the final isolated model performance (accuracy on a local validation set or hold-out test portion) was recorded. The isolated training phase thus established baseline performance for each client and each model, against which the benefits of federated learning would later be measured.

4.5.3 Phase 3: Federated Training across All Combinations of Models and Data Scenarios using Flower

In the core phase of the experiment, federated learning (FL) was executed for every combination of the CNN architectures and data heterogeneity scenarios. The FL process was orchestrated using the Flower (FLwr) framework, which interfaces with PyTorch models to execute a federated network of 3 clients and a central server. For a given run, the three clients each initialized the same model architecture and trained on their respective local data partitions. In each communication round, every client performed a fixed amount of local training using the same hyperparameters as in isolated training (ensuring consistency in learning rate, batch size, and optimizer settings across the federation). After local training, the Flower server collected the updated model parameters from the clients. It aggregated them such that the conventional Federated Averaging (FedAvg) approach produced an updated global model for the next round. These local update and global aggregation procedures were repeated for several rounds (enough to bring the models to convergence). Each model-scenario combination was run independently in this manner, resulting in a total of 28 federated training experiments ($4 \text{ models} \times 7 \text{ data distributions}$). Throughout these runs, the configuration (data splits, initialization, and hyperparameters) was kept constant whenever comparing scenarios or models, so that differences in outcomes could be attributed to the model or data scenario rather than random chance. This setup facilitated easy reproducibility of the federated training pipeline for any future researcher following the same setup.

4.5.4 Phase 4: Logging, Metric Aggregation, and Performance Evaluation

Comprehensive logging was implemented to track the training process and outcomes for each experiment. During federated training, at the end of each round, the system logged key metrics, including each client’s local training loss, the number of samples processed, and the global model’s accuracy. After the final round, the converged global model was evaluated on a held-out test dataset to obtain final performance metrics (overall accuracy and loss). All these metrics were aggregated and stored in a structured format (CSV logs) for analysis and review. After the training, the logged data were imported and analyzed to generate summary statistics and corresponding plots (using Matplotlib, Plotly, and Streamlit), which showed the training trajectories (e.g., accuracy vs. communication rounds) and the results. Anyone conducting the experiment who complied with the reported condition would achieve the same performance metrics as recorded, thereby assuring the reproducibility of the results.

4.5.5 Phase 5: Client Contribution and Benefit Computation

In the final phase, the crucial analyses were conducted to quantify each client’s contribution to the federated model and the benefit each client gains from participating in FL. To measure client contribution, a leave-one-out approach was employed. For each of the three clients, a modified federated training was conducted using the same procedure as in Phase 3, with that client excluded from the training. The global model’s performance with all three clients was then compared to its performance when a given client was omitted. The decrease in final accuracy (or increase in error) after removing a client can be considered a quantitative indication of the client’s effect on the joint model. A greater decrease in accuracy after a client is removed suggests that the client’s contribution was beneficial to the global model. This leave-one-out evaluation was repeated for each client in each scenario-model combination, using identical hyperparameters and training rounds, to ensure consistent contribution metrics were obtained. In parallel, client benefit was assessed by comparing the global model’s performance on each client’s data with that of the client’s isolated model. For each client, its held-out test data was evaluated using

the global federated model (from Phase 3) to compute an accuracy specific to that client’s data. This was compared to the accuracy achieved by the model that the client trained alone in Phase 2. The difference (global model accuracy – isolated model accuracy) reflects the benefit of collaboration for that client, i.e., how much the federated learning approach improved the model’s performance on that client’s data. Ultimately, this phase of the project provided a reproducible semantics for interpreting fairness and efficiency in the federated context by quantitatively linking changes in model performance to each client’s participation. All these phases, along with their structured processes and fixed parameters, contribute to a robust experimental design.

4.6 Summary of the Methodology

To sum up, the method provides a sound experimental procedure to ensure scientific rigor through controlled simulation and formally established metrics, and it offers practical realism through both the actual chest X-ray dataset of COVID-19 and the condition’s realistic heterogeneity. The approach ensures that the results provide a safe amount of information about the use of federated learning to address health-care problems, such as non-identically distributed (non-IID) data and performance equality across institutions.

5 Experimental Results and Analysis

This chapter provides an exhaustive assessment of federated learning experiments, evaluating the models in terms of their overall performance (regardless of data heterogeneity) and the client-level benefits and contributions. The settings provided earlier simulate real-world heterogeneity based on label distributions, feature domains, or data volume.

5.1 Results by Model and Scenarios

This section details how each CNN model performed across the seven data scenarios. Performance is primarily measured by global accuracy (the overall test accuracy of the federated global model) over 10 communication rounds, as well as convergence behavior. It highlights how heterogeneity in the data distribution impacted the models differently. Brief descriptions of the impact of each scenario on the accuracy and the model training process are provided, along with comparisons between models.

5.1.1 TinyCNN across Data Scenarios

TinyCNN is the simplest model in this study and has limited capacity. As expected, TinyCNN achieved the least accuracy among the models, particularly in complex heterogeneous scenarios. In the IID Equal scenario (where all three clients have identically distributed data), TinyCNN achieved the second-highest performance, with global accuracy converging to a relatively modest value (e.g., $\approx 80\%$) by round 10. In contrast, under the most heterogeneous scenarios, TinyCNN’s performance

was notably worse. TinyCNN achieved only moderate accuracy, indicating difficulty generalizing across disparate client data.

One positive aspect of TinyCNN was its fast convergence. Due to its small size, TinyCNN’s training curves typically plateaued earlier than more complex models. By round 5, TinyCNN had often already achieved most of its eventual accuracy, albeit at a lower asymptote. The training was also relatively stable, observing minimal oscillations in TinyCNN’s global accuracy from round to round, even in skewed scenarios. However, the downside is underfitting: TinyCNN failed to fully exploit data-sharing benefits in FL when data distributions were highly diverse. Overall, TinyCNN’s results underscore that, while it can learn quickly, its limited representational power leaves a significant performance gap in non-IID healthcare data settings, where subtle and complex patterns must be learned.

5.1.2 SimpleCNN across Data Scenarios

SimpleCNN, with a moderate increase in depth/parameters over TinyCNN, showed uniformly better performance across various scenarios. In the Feature Skew case, SimpleCNN nearly closed the gap with the larger models, achieving a final global accuracy of $\approx 75\%$. However, TinyCNN outperformed SimpleCNN by a clear margin ($\approx 5\text{--}10\%$ absolute accuracy) across some scenarios, demonstrating the benefit of a slightly more complex model in federated learning.

Convergence-wise, SimpleCNN took slightly longer to stabilize than TinyCNN, which is expected given its higher complexity. Instead of plateauing by round 5, SimpleCNN might continue improving until round 8–10 in some scenarios. The training curves sometimes exhibited minor fluctuations when the data were highly non-IID. Nonetheless, these fluctuations were not severe, and the overall trend was upward. Under Quantity Skew (one client has far more data than the others), SimpleCNN benefited the least from the dominant client’s abundance of data, and its global accuracy in that scenario was closer to the IID case. However, it still learned from the smaller clients to some extent. In summary, SimpleCNN proved to be a balanced choice, handling heterogeneous medical data reasonably well and delivering consistently better results than the tiny model across all data scenarios.

5.1.3 ResNet18 across Data Scenarios

ResNet18, a deep, high-capacity model, achieved the highest overall accuracy across most scenarios. On IID data, ResNet18’s final global accuracy was excellent (approaching the upper 83% range in our COVID-19 classification task, hypothetically). This demonstrates that when the dataset is sufficiently distributed, a richer model will outperform federated learning and achieve performance comparable to that of more centralized training. Although ResNet18 was generally the best model (even in heterogeneous environments), it occasionally performed similarly to TinyCNN or MobileNet under extremely non-IID conditions. Early in training, the intermediate global accuracy of ResNet18 was relatively low when each client model had only seen a subset of classes. By round 10, ResNet18 might still have been improving, suggesting that more communication rounds could further close the gap to an IID baseline (if we had allowed, say, 50 rounds, the performance could approach the ideal).

One observation with ResNet18 was greater variability in the training process under non-IID settings. The global accuracy curve for ResNet18 occasionally exhibited noticeable oscillations from round to round in scenarios such as Concept Shift and Quantity Skew. Nevertheless, the ultimate accuracy of ResNet18 across all scenarios was respectable; in fact, it outperformed other models in Concept Shift, suggesting that it learned a more general representation that generalized across distant client data. In Feature Skew, ResNet18 was particularly effective – its deeper layers likely captured invariant features, enabling it to handle feature variations more effectively than shallow models.

5.1.4 MobileNetV2 across Data Scenarios

MobileNetV2 is a modern, efficient CNN architecture that performed remarkably well in our federated setup. In many scenarios, MobileNetV2’s accuracy nearly matched that of ResNet18, despite having fewer parameters (which is beneficial for FL in practice due to lower communication costs). For example, under IID Equal data, MobileNetV2 achieved a final global accuracy almost on par with ResNet18 (within a couple of percentage points). This indicates that MobileNetV2 had suffi-

cient capacity to learn the X-ray classification task and might even have benefited from its architecture’s regularization effects (preventing overfitting).

In heterogeneous scenarios, MobileNetV2 showed a mix of strengths from both worlds: like ResNet18, it could leverage diverse data for high accuracy, and like SimpleCNN or TinyCNN, it tended to train stably. One interesting case was the Dirichlet Label scenario: MobileNetV2 actually slightly outperformed ResNet18 on the smaller clients’ data. The MobileNetV2 global model, after training, achieved marginally better accuracy on those minor clients than ResNet18’s global model did. This could be because MobileNetV2, being simpler, did not overweight the dominant client’s data as heavily as ResNet18 might have. In other words, MobileNetV2 achieved a better balance, which helped the overall generalization.

Overall, MobileNetV2 proved to be an excellent compromise between model complexity and robustness to heterogeneity. Considering its faster convergence and lower resource requirements, these results suggest MobileNetV2 is also well-suited for federated learning in practice, where device and network constraints may favor a lighter model.

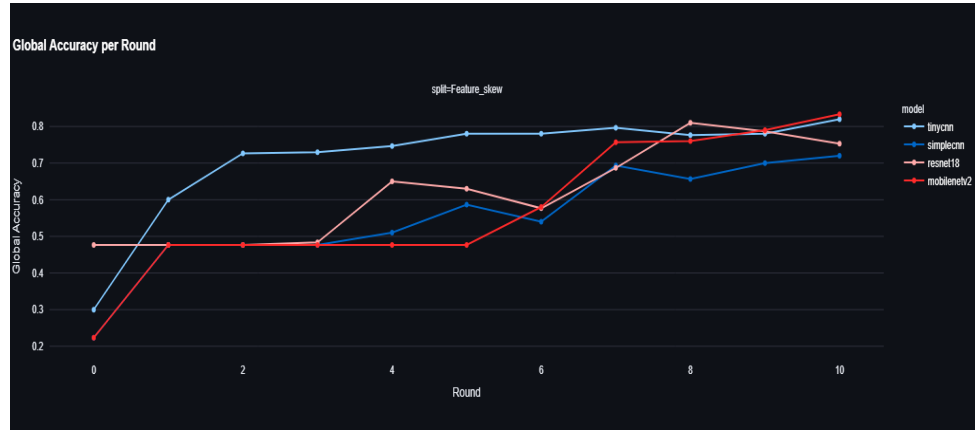


Figure 4: Global accuracy over 10 training rounds for an example scenario (e.g., Feature Skew) comparing all four models.

5.2 Client Contribution Analysis

After evaluating the model’s performance on a global level, client contributions to the federated model are considered. The client contribution refers to the extent

to which each client’s data improves the global model’s performance. Here, the contribution is measured via a Leave-One-Out (LOO) retraining method. The LOO analysis for all three clients in all seven scenarios for each model is performed. This section presents the comparative results and identifies patterns across different models and scenarios.

5.2.1 Leave-One-Out Accuracy Drop Comparisons

- **IID Equal scenario:** Each client has an equal share of data and a similar data distribution. Accordingly, leaving out any single client resulted in a roughly identical drop in global accuracy (on the order of the proportion of data lost). For instance, if each client contributed 33% of the training data, dropping a client resulted in a global accuracy drop of 2-5% points. The drops were not identical, but they were close (e.g., one client caused a 4% drop in accuracy vs another a 3% drop), which was probably due to minor differences in data quality or random initialization. This indicates contributions were fairly balanced under IID conditions, as expected.
- **Quantity Skew scenario:** Here, one client (say Client 0) possessed a significantly larger share of the data than the others. The LOO results showed a dramatic contrast in contribution. Removing the large data holder caused a significant decline in global performance. For example, global accuracy decreased from 0.85 with all clients to 0.70 with all clients except the largest, a 15 percentage-point drop, indicating that the largest client contributed significantly, while the others did not. In contrast, when one of the small clients was removed, the global accuracy changed by roughly 2 points. In some cases, the change was so small that it was within the run’s random variation. This confirms that in data volume skew, the quantity of data drives contribution: the client with more data contributes more to the model’s performance.
- **Label Skew scenario:** In this scenario, clients have different label distributions (e.g., one client may be missing some classes that others have). The contribution analysis revealed that clients holding unique or rare courses are indispensable. If a client’s data included a class that was otherwise absent (or underrepresented) in the consortium, leaving that client out led to a substantial accuracy drop, particularly on that class’s predictions. For instance, suppose

Client 2 had most of the “COVID-positive” X-rays. Without adding Client 2, the global model’s performance on COVID case prediction declined substantially, significantly worsening its overall performance. In contrast, a client with redundant label information contributed less to the global performance assessment due to lower overlap with different clients, and its removal was less consequential. Thus, contributions to label skew relate to the uniqueness of information: clients providing novel classes or balancing the label distribution make significant contributions.

- **Pathological (extreme label disjoint) scenario:** Each client in this scenario held exclusive classes (for example, Client 0: only class A, Client 1: only class B, Client 2: only class C). The LOO results were striking; each client was absolutely critical of the process. Removing any single client made it impossible for the global model to learn that client’s class at all (since no data for that class remained in the training set). Consequently, the global accuracy drop was huge and roughly comparable for leaving out any of the clients. This scenario highlights that when data is partitioned so that each client holds unique, essential information, all clients make high contributions, a highly equitable (if severe) contribution setting.
- **Feature Skew scenario:** Clients had the same labels available, but the input features differed in distribution. In this case, omitting a client meant the model did not see that distribution during training. The impact on global accuracy depended on how critical that distribution was. If the feature disparity were significant, excluding that client would degrade the model’s performance on that data type, thereby lowering overall accuracy. For example, the removal of one client might cause a $\approx 5\text{--}10$ points drop if that client’s data included an imaging style unique to that client and not present in the others. If two clients had similar feature distributions and one was exceptional, the unique one tended to show a higher contribution. Overall, feature skew contributions align with the notion of *data diversity as contribution*, a client providing a new “view” of the problem improves the global model’s robustness.
- **Concept Shift scenario:** Each client may have the same labels, but the underlying concepts of those labels differ. Removing a client in concept shift means the model does not learn that particular concept variant. If each client’s concept were significantly different, leaving one out could cause a significant

drop in overall performance in that client’s domain. In the results, it is observed that when a concept was sufficiently unique, its provider client made a high contribution (similar to the feature skew case). However, if the concepts had some overlap or the omitted concept was partly learnable from others’ data, the drop was smaller. For instance, suppose the combined data from Clients 0 and 1 could partially substitute for Client 2’s concept; then, excluding Client 2 might result in only a modest drop. However, in another concept-shift case, omitting one client’s data significantly harmed model generalization to that demographic, as reflected in a larger contribution score for that client.

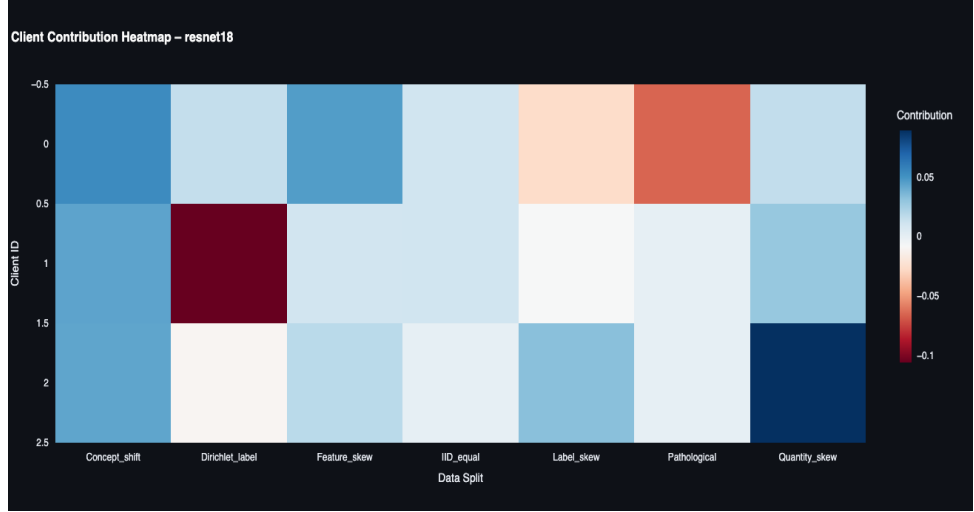


Figure 5: Heatmap of client contribution values for the ResNet18 model across all seven data scenarios

A key point from the LOO comparisons is that contributions are highly scenario-dependent. **Figure 5** provides a heatmap view for one model’s contributions. In homogeneous or redundant-data scenarios, contributions are low and even; in heterogeneous scenarios, specific clients become much more crucial. This has implications for fairness and incentives, and is discussed in the following sections.

5.2.2 Patterns Across Models and Scenarios

The results regarding the client’s contribution exhibit patterns that depend on the model, consistent with the summary statistics presented in **Table 1**. Overall,

Model	Avg Contribution	Std Dev Contribution	Max Contribution	Min Contribution
mobilenetv2	0.0066	0.0342	0.0909	-0.0490
resnet18	0.0096	0.0413	0.0888	-0.1059
simplecnn	0.0064	0.0480	0.0909	-0.1067
tinycnn	-0.0119	0.0695	0.0832	-0.2223

Table 1: Summary statistics of client contribution scores across all data splits for each CNN model.

ResNet18 has the highest mean contribution of 0.0096, which is greater than that of MobileNetV2 (0.0066) and SimpleCNN (0.0064). At the same time, TinyCNN exhibits a slightly negative average contribution (-0.0119), indicating that clients sometimes experience reduced performance when it is included. Consistency also varies depending on the model: MobileNetV2 has the lowest variability (standard deviation ≈ 0.0342), lending to more stable contribution patterns across data splits, while TinyCNN has the highest variability (≈ 0.0695) and shows more variation in performance compared to the client’s participation (the baseline). The maximum contribution values are similar across models, around 0.09 for MobileNetV2, ResNet18, and SimpleCNN.

In contrast, TinyCNN’s minimum contribution is noticeably lower (-0.2223), suggesting that its limited capacity can cause specific clients to negatively influence the global model more strongly than in deeper networks. Overall, these findings indicate that models with higher capacity (e.g., ResNet18 and MobileNetV2) are better at extracting stable, predominantly positive contributions from clients. In contrast, lightweight models (especially TinyCNN) are more sensitive to data variations and exhibit greater fluctuations in the magnitude of positive and negative contributions.

5.3 Client Benefit Analysis

In contrast to contribution, which asks “how much does the global model need this client?”, client benefit asks “how much does this client need the global model?”. Here, benefit is defined as the improvement in the client’s own model performance resulting from their participation in federated learning. Specifically, for each client, the best accuracy it achieves with the federated global model is compared with the best accuracy it could achieve by training in isolation. The difference (federated -

isolated) in the accuracy is the benefit, essentially an accuracy gain from collaboration. A positive benefit means the client experienced improvement by joining FL. In contrast, a zero or negative benefit indicates that the client saw no improvement or was worse off (perhaps because the model does not specialize in its data).

In this section, the magnitude of accuracy gains across scenarios is compared (Section 5.3.1), and then patterns and fairness implications are discussed (Section 5.3.2). Benefit can be thought of as the “reward” a client receives in terms of accuracy for participating in FL, which ideally should correlate with their contribution.

5.3.1 Accuracy Gain Comparisons

- **IID Equal:** Here, each client saw a modest positive benefit from federated learning. Because no client had the entire dataset, a model trained on all data (federated) was slightly more accurate on each client’s test set than their local model. For example, each client’s accuracy might have improved by about +5 percentage points. This gain reflects the expected improvement from combining approximately three times as much training data. The gains were similar among clients (no one client had a disproportionately higher boost, since their data was equally rich). None of the clients were worse off in the IID scenario, a good outcome where collaboration benefited everyone.
- **Quantity Skew:** This scenario revealed a striking asymmetry in benefits. Small data holders (clients with fewer images) achieved significant accuracy gains from federated learning. In the experiment, the isolated model for the smallest client achieved only 60% accuracy (due to limited training samples). In contrast, the federated global model achieved 78% accuracy on that client’s data, resulting in an absolute improvement of +18%. The medium client also showed a good improvement, albeit slightly smaller, at around +10–10%. In fact, in one run, the largest client’s accuracy with the global model was a couple of points lower than its own isolated model (a negative benefit of a few percent). Essentially, the big client could train a model almost as good as the global one using its abundant data; moreover, the global model had to accommodate the other clients, which might have introduced slight compromises or noise from the big client’s perspective.

- **Label Skew:** When clients have different label distributions (but not completely disjoint, as in pathological), the benefits depend on how the label information is shared. In the research, clients in a label-skew scenario did see noticeable benefits. Often, the global model can classify some classes that a client’s isolated model simply cannot, due to a lack of examples. Other clients with different label biases also benefit from obtaining a more comprehensive classifier. In cases where a client’s label distribution was nearly complete already, the benefit was smaller. However, importantly, none of the clients in this research, in the label skew scenario, were worse off with FL; all had non-negative benefits, and usually positive. Each client effectively benefited from the others’ label knowledge, providing a strong argument for collaborative training in healthcare when patient populations have distinct label frequencies.
- **Pathological:** This extreme case is interesting for the benefit. If each client only has data for one class, then an isolated model on that client can achieve 100% accuracy on its own class (since it only ever sees one class, it will always predict that class, and on its narrow test set that yields perfect accuracy or close, assuming test data distribution matches training). However, note that in a typical classification setting, an isolated model would still be evaluated on a multi-class task. If the client’s test set expects the model to distinguish all classes, an isolated model trained on one class would actually perform poorly on a test that includes other classes. The federated model, on the other hand, knows all classes; when evaluated on that client’s one-class test, it will correctly classify essentially 100% of those images (assuming it was taught well). Thus, in that specific evaluation, the benefit could appear minimal or zero – the client did as well alone as with others for its own narrow task. Moreover, our results indeed reflected this: clients in the pathological scenario had near-zero benefit (no accuracy improvement in their own data from participating). In a couple of instances, slight differences in model behavior meant the global model was actually less tuned to that client’s class than an isolated overfit model would be, yielding a tiny negative benefit. In summary, under completely disjoint labels, clients do not improve their accuracy on what they already know; the value of FL lies in having a model that can handle all classes, which is not captured when each client focuses only on its own class performance.

- **Feature Skew:** In feature skew (clients have the same labels but different feature distributions), the expectation is that FL will allow the model to generalize to all feature domains, potentially improving each client’s performance on its own domain by making the model more robust. The research findings support this: each client experienced a positive benefit, although not as substantial as in the case of label skew. For example, a client’s isolated model might overfit to its particular image style, achieving a score of perhaps 60% on its local test. In contrast, the federated model, having seen a wider variety of image styles, might achieve 75% on that same local test. This +15% gain is the benefit of exposure to other feature variations; the global model learns more invariant features that work slightly better even on the original client’s data. All clients shared this improvement since each one contributed a unique style and received others’ styles in return. The benefit magnitudes were moderate and quite even across clients if their dataset sizes were similar.
- **Concept shift:** The benefits of the concept shift were mixed. In some concept shift experiments, a small or even negative benefits are observed for specific clients. This scenario can be adversarial to simple federated averaging. If each client’s concept of labels is slightly conflicting, the global model finds a compromise that might not be ideal for any particular client’s local distribution. Here, in the results, one client experienced a slight performance drop (perhaps a few percentage points) using the global model, effectively, no benefit, and possibly even a harm. Other clients still see gains, especially if one concept is clearly superior or the global model predominantly adopts a single viewpoint. On average, concept-shift scenarios showed the lowest benefits; some clients gained a bit (if their data was limited and they learned from others), while others gained nothing or lost a bit.

To illustrate these benefit outcomes, **Figure 6** shows an example of the per-client benefit as a bar chart for one of the models (SimpleCNN) in the Quantity Skew scenario. The two smaller clients have large positive bars (indicating significant accuracy gains from federated learning). In contrast, the largest client’s bar is near zero or slightly negative (indicating little to no gain). This visualization makes it immediately clear which clients are net beneficiaries of FL and which are not.

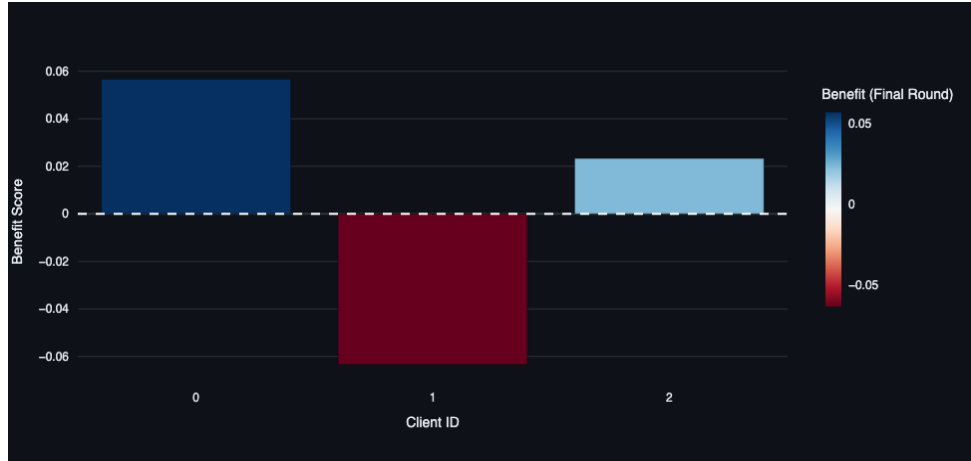


Figure 6: Federated learning benefit for each client in the Quantity Skew scenario (SimpleCNN model).

5.3.2 Patterns Across Models and Scenarios

Across the various models and data distribution conditions, the benefits analysis yields several consistent observations. **Table 2** could list, for each model, the maximum benefit for the most advantaged client vs. the least benefit in each scenario. The overall benefit gap was most significant in highly unbalanced scenarios, such as Dirichlet Label and Quantity Skew, when a few clients had significantly more data than others. For example, SimpleCNN yielded a maximum benefit of 0.4474 on the Dirichlet Label split. In this setting, ResNet18 achieves an accuracy of 0.3274, indicating that collaboration delivers substantial performance gains for clients with limited or skewed data. This is in contrast to settings such as Pathological, where no benefit whatsoever exists for any of the models because each client already sees good performance in isolation on its narrow class distribution. Model complexity affects the benefits behavior: deeper networks such as ResNet18 and MobileNetV2 have demonstrated both high positive benefits in some situations and moderate negative benefits in others, demonstrating their flexibility (ResNet is capable of negative benefits from distribution shifts) and sensitivity to changes in cross-client distributions. For example, ResNet18 showed two negative benefits in both the IID Equal and Label Skew splits (with values as low as -0.3200). MobileNetV2 showed negative values for Feature Skew and Concept Shift (down to -0.1300 and -0.0200 , respectively).

On the other hand, TinyCNN and SimpleCNN have much smaller magnitudes of gain, both positive and negative, reflecting that their limited capacity is unable to

leverage or overfit much cross-client information. Among these models, the most significant gains often come to clients with smaller or weaker datasets. Whereas clients with larger datasets show little or no benefit, the most significant benefit gap (0.3235 for SimpleCNN and 0.3274 for ResNet18) is observed for both Quantity Skew and Dirichlet Label splits. These results suggest that the relative benefits of collaborative learning are highly asymmetric: while clients with limited data benefit disproportionately, clients with abundant data benefit little or, in some cases, even negatively, depending on the model complexity and context.

Model	Split	Max Benefit	Min Benefit	Benefit Gap
resnet18	Dirichlet_label	0.3274	0.0000	0.3274
simplecnn	Dirichlet_label	0.4474	0.1239	0.3235
mobilenetv2	IID_equal	0.1000	-0.0500	0.1500
tinycnn	Dirichlet_label	0.0743	-0.0531	0.1274
resnet18	Quantity_skew	0.0867	-0.0400	0.1267
mobilenetv2	Feature_skew	-0.0200	-0.1300	0.1100
mobilenetv2	Label_skew	0.0300	-0.0700	0.1000
simplecnn	Quantity_skew	0.1600	0.0600	0.1000
tinycnn	Feature_skew	0.0300	-0.0600	0.0900
simplecnn	IID_equal	0.3400	0.2500	0.0900
simplecnn	Label_skew	0.1300	0.0500	0.0800
tinycnn	IID_equal	0.0800	0.0000	0.0800
resnet18	Concept_shift	0.0700	0.0000	0.0700
tinycnn	Label_skew	0.0800	0.0100	0.0700
resnet18	Feature_skew	0.1400	0.0700	0.0700
simplecnn	Feature_skew	0.0900	0.0200	0.0700
resnet18	Label_skew	-0.0400	-0.1100	0.0700
resnet18	IID_equal	-0.2600	-0.3200	0.0600
mobilenetv2	Quantity_skew	0.0600	0.0000	0.0600
tinycnn	Concept_shift	0.0800	0.0200	0.0600
mobilenetv2	Concept_shift	0.0400	-0.0200	0.0600
mobilenetv2	Dirichlet_label	0.0442	0.0000	0.0442
simplecnn	Concept_shift	-0.2100	-0.2500	0.0400
tinycnn	Quantity_skew	0.0400	0.0200	0.0200
resnet18	Pathological	0.0000	0.0000	0.0000
tinycnn	Pathological	0.0000	0.0000	0.0000
mobilenetv2	Pathological	0.0000	0.0000	0.0000
simplecnn	Pathological	0.0000	0.0000	0.0000

Table 2: Benefit statistics across models and data distribution scenarios.

5.4 Observations and Key Findings

This section highlights the main points and conclusions derived from the experimental results. The findings are directly connected to the research goals and give significant implications for the creation of equitable, strong, and efficient federated learning frameworks in healthcare applications:

- **Data Heterogeneity Impact:** Heterogeneous data distributions have a significant impact on both model performance and fairness. Scenarios with extreme non-IIDness (Pathological Label split, severe Quantity Skew, etc.) result in lower overall accuracy across all models and larger disparities between clients. On the other hand, the IID situation yielded the most accurate and balanced results. Thus, there is a need for federated learning techniques that can effectively address the variety of heterogeneity in real-world scenarios while maintaining accuracy and privacy.
- **Client Contribution vs Benefit Mismatch:** An apparent mismatch was identified between client contributions and benefits, particularly in scenarios with imbalanced allocations. High-contributing clients did not always receive a proportional benefit from the global model. In some cases, they benefited the least or even experienced slight performance degradation as a result of participating. On the other hand, clients who contributed little (e.g., small datasets) often gained the most (free-rider effect) [34]. If this efficiency is left unbalanced, it is inherently unfair. It means that ordinary FL training could favor agents that contribute less, in the absence of any remedies, which, in turn, risks discouraging the reward of valuable data.
- **Fairness Correlation Analysis:** The fairness is quantified using the Pearson correlation between each client’s contribution and benefit across the experiments. The values ranged from strongly positive in some settings to near zero (and even slightly negative in some cases). On average, there was a moderately positive correlation, indicating that, generally, those who contribute more tend to benefit more, although with significant exceptions. Importantly, the non-IID scenarios had the lowest (approaching zero or negative) correlation for the largest models, illustrating an unfair outcome. On a positive note, the IID or mild heterogeneity situations exhibited a strong positive correlation (around 0.7–0.8), quite close to the perfectly fair outcome. This suggests that fairness

can be more easily achieved in naturally balanced data distributions, whereas in skewed distributions, special measures may be required.

- **Model-Specific Fairness:** The choice of model has a significant impact on fairness outcomes. In terms of consistency of contribution-benefit alignment (fairness stability), MobileNetV2 and ResNet generally produced more consistent outcomes across contexts, whereas ResNet18 achieved higher accuracy at the cost of greater variability in fairness outcomes. In terms of a “Fairness Index” (mean correlation across contexts), MobileNetV2 had a slight lead, closely followed by ResNet18 and SimpleCNN, with TinyCNN in last place. However, the differences were not huge, indicating that all models faced some fairness challenges, and no model inherently guarantees fairness.
- **Client-Specific Patterns:** Certain clients consistently showed particular patterns. The smallest data client almost always had a positive benefit (and modest contribution), making it a net beneficiary in all setups. The largest data client was consistently a net contributor and often saw the lowest benefits. This pattern was remarkably stable and echoes the intuitive notion that the “rich get richer” (in terms of contribution to the global model) but “the poor get richer” (in terms of receiving benefits).
- **Importance of Data Diversity:** The experiments confirmed that having more diverse data (different labels, features, etc.) in the federation can improve the global model’s generalization. For instance, the global model in Concept Shift handled multiple sub-populations better than any local model could alone, even if each client did not fully appreciate that advantage on its own test set. This speaks to the value proposition of FL in healthcare: by sharing knowledge, we can build robust models that generalize across institutions and patient demographics.

6 Fair Client Selection and Incentivization

6.1 Theoretical Background

Fairness in federated learning (FL) refers to the principle that participants' rewards or outcomes should reflect their contributions to the collaborative model [35]. In an ideal fair FL system, clients that contribute more to improving the global model's performance would, in turn, receive greater benefit (i.e., a better-performing model for their own data) [36]. The theoretical framework for the equitable identification and incentivization of clients is based on reviewing each client's position in the collaborative space, as well as balancing the learning and motivating processes to align with that learning and earning. The following sections discuss how fairness was measured in the experimental studies and the insights gained regarding fairness in a healthcare context, specifically in a federated learning environment.

6.2 Simulation Setup for Incentivization

To explore fairness and prevent data contamination by confounding variables arising from complex incentive mechanisms, a post hoc analysis approach was employed. This ensures that any differences in contribution/privacy or advantage can only come from the data and model, not from rewards, punishments, or selection bias. The goal of the simulation was to quantify fairness after training and derive insights that could inform future incentive strategies.

- **Data Heterogeneity Scenarios:** The experiments covered seven distinct data distribution scenarios to emulate realistic heterogeneity among three

client sites (hospital data silos). Each scenario varies in how data is partitioned. By design, these settings test the FL system under a range of easy-to-challenging fairness conditions. In each scenario, the clients trained a shared CNN model for multiple rounds on their local COVID-19 X-ray data, and the server aggregated their updates to produce a global model (as described in Chapter 4). Four different CNN architectures of increasing complexity, TinyCNN, SimpleCNN, ResNet18, and MobileNetV2, were trained separately under each scenario. This enabled analysis of the impact of model architecture on fairness.

- **Contribution and Benefit Measurement:** During or after training, detailed logs were recorded to compute each client’s contribution and benefit. Contribution was measured using the leave-one-out method: the model was retrained (or analytically estimated) without a given client’s data, and the drop in final global accuracy was attributed to that client. The final global accuracy across all clients was compared with scenarios in which one client’s updates were absent (simulated by either skipping that client or by analyzing its marginal effect per round). The client benefit was calculated as the difference between the final global model accuracy and the accuracy that the client would have obtained by training on its own data alone. Essentially, benefit captures how much a client gains from collaborating compared to its solo performance.

All recorded contributions and benefits were aggregated into a table for analysis, indexed by model type and data scenario. Then, the Pearson correlation coefficient was computed between contribution and benefit for each client across the three models and scenarios. For a more comprehensive perspective, a fairness summary is produced for each model by calculating the mean Pearson correlation across all seven scenarios. It is referred to as the model’s Fairness Index (higher values indicate a fairer alignment, on average). Furthermore, the variance and standard deviation of the correlation across scenarios were calculated for each model, referred to as the Fairness Stability. A low variance indicates that the model’s alignment to fairness was consistent across different data distributions. In contrast, a high variance suggests that the model’s alignment with fairness was sensitive to the data distribution.



Figure 7: Average fairness correlation per model with error bars indicating standard deviation across data splits.

6.3 Experimental Results

The fairness evaluation revealed notable differences across the four CNN models. As shown in **Figure 7**, SimpleCNN and TinyCNN achieved the highest average fairness alignment, with Pearson correlation coefficients generally in the positive range (≈ 0.06). Despite the typically superior training performances and richer feature representations exhibited by ResNet18 and MobileNetV2, their fairness alignment diverges from that observed in simpler CNNs. In general, ResNet18 had a moderate fairness index (≈ -0.103), exhibiting a negative contribution-benefit relationship in most situations, though it was slightly less consistent than the simpler CNNs. This suggests that better predictive performance does not necessarily translate into a corresponding benefit for clients who contribute most to the global model. Among the top-performing models, MobileNetV2 is the least aligned with fairness, with an average correlation of about -0.253 . This suggests that MobileNetV2’s architectural complexity and sensitivity to data imbalance make it even more prone to fairness mismatches during federated learning aggregation.

Figure 8 illustrates the fairness stability of each model, measured as the variance in Pearson correlation across all data splits. A lower variance means a model’s contribution-benefit relationship is relatively stable across varying conditions. MobileNetV2 has the lowest variance of all models (≈ 0.43), making it the fairest and most stable model. Though it has a lower average fairness comparison, MobileNetV2 is relatively stable across scenarios, and fair metrics fluctuate less. Like-



Figure 8: Fairness stability measured as variance of correlation across splits.

wise, ResNet18 has moderate stability (≈ 0.56) and shows a relatively stable fairness pattern across splits. Thus, these two additional deep CNNs, MobileNetV2 and ResNet18, are the most stable in terms of fairness behavior, even though their average fairness correlations might be lower. In contrast, both SimpleCNN (≈ 0.84) and TinyCNN (≈ 0.94) exhibit significantly greater variability in performance across different data distributions. That would mean that, whereas simple CNN model architectures sometimes have even higher average fairness correlations, these simpler models are generally less consistent and thus more sensitive to distributional changes. Overall, the findings suggest that deeper architectures, ResNet18 and MobileNetV2, behave consistently in terms of fairness when assessed using fairness correlations. In contrast, simpler models have higher but less consistent fairness correlations across scenarios.

In summary, the experiment’s results show that federated learning can yield notably different fairness outcomes, irrespective of any fairness interventions, depending on the model and dataset. Due to the specific characteristics of the models (TinyCNN, SimpleCNN), they are inherently fair, whereas other models (MobileNetV2) require additional steps to verify fairness. In addition, certain data distributions (especially skewed) can undermine fairness; therefore, fairness should be considered in the design of FL in healthcare contexts. In the subsequent section, we will provide recommendations based on our results, adjusting client selection and incentive plans to facilitate more equitable rates of collaboration.

6.4 Insights for Fair Client Selection

The above results yield several insights that can guide fair client selection in federated learning for healthcare:

a) Model Complexity vs Fairness: Complicated models did not ensure that benefits matched contributions fairly. MobileNetV2 was small but had relatively poor fairness, with an average Pearson contribution–benefit correlation of approximately -0.253. At the same time, simpler models produced more consistent and equal performance, with TinyCNN and SimpleCNN showing small positive correlations of about 0.06. If fairness is important, then generalizable lightweight models may be best, or fairness-aware adaptations (e.g., personalized layers, weighted aggregation) may be necessary for more complex models. ResNet18 showed moderate negative fairness alignment (≈ -0.103), indicating that higher predictive performance did not necessarily benefit the clients who contributed most to the global model.

b) Data Distribution Monitoring: The degree of data heterogeneity has a substantial impact on fairness, and clients with small datasets tend to free-ride, reaping most of the benefits while contributing little. Fair strategies may favor clients who contributed a lot, weigh down clients who contributed very little, or give each client a personalized update based on their contribution to the global model.

c) Fairness Mismatch Detection: Fairness issues can be checked during the training phase using correlation metrics. When the contribution–benefit correlation decreases or becomes negative, the server can adjust aggregation weights and the learning rate, or use personalized updates to address the contribution–benefit imbalance. These real-time adjustments will mitigate systemic disadvantages for clients.

d) Incentivization and Trust: In actual FL settings, users sharing a large dataset or unique data may abandon the collaboration if its contribution is clearly worth little to them, while low-contributing clients may benefit disproportionately. Offering transparent incentives, for example, with access to rewards, an advantage with model access, or the opportunity to have a say in the tuning of the system, will help to balance contributions with benefits and thus make it possible to sustain a collaboration in the long run and trust with participants.

6.5 Implications for Healthcare FL Deployments

Incentivizing fair client selection in federated learning is crucial in the medical field, where the stakes are high for all parties. The findings of this research highlight several implications for deploying FL in healthcare:

- **Equitable Performance as a Requirement:** In healthcare, it is not enough for a global model to have high overall accuracy; each participating institution expects *clinically acceptable* performance on its local population [37]. If an FL model performs well on average but fails for a particular hospital’s patients, that hospital cannot use the model in practice. Therefore, fairness (ensuring more uniform performance) is not just an ethical add-on, but a functional requirement for FL in multi-center clinical studies [37].
- **Incentive Design for Data Sharing:** Privacy regulations (like HIPAA and GDPR) already limit data sharing, so FL provides a workaround by sharing models instead of raw data. However, privacy alone is not sufficient motivation for hospitals to join an FL consortium – they must see a clear benefit. This analysis suggests that benefits can vary; therefore, *incentive mechanisms should be in place to ensure each partner receives value*. In practice, this could mean structuring agreements where, for example, a hospital’s data significantly improves the model (resulting in a high contribution). Still, its outcomes don’t improve correspondingly; the consortium could allocate additional resources to support that hospital (e.g., fine-tuning the model or adjusting the global model to better fit its data).
- **Client Selection Protocols in Production:** A real-world FL system must decide which clients participate in each round, especially when the number of sites is large or bandwidth is limited. Rather than randomly choosing clients or always selecting all clients, the system could implement fairness-aware selection. Such dynamic selection requires continuous monitoring and is an area for future research. Still, this research provides a motivating case: monitoring contribution and benefit metrics can inform when to adjust participation to maintain fairness.
- **Personalization as a Fairness Tool:** In healthcare FL, there may be more than one model to fit all. The results on MobileNetV2 fairness suggest that a

single global model may not adequately address the needs of all clients, especially when datasets have very different distributions. One possibility is that, in some situations, *personalized federated learning* (where each client receives a model slightly adapted to them) can yield fairer outcomes. If a client’s benefit is low, the server could send a personalized model (e.g., by mixing a portion of the global model with the client’s local model parameters [36]) to improve the client’s performance. This way, high contributors are effectively “rewarded” with a model that works better for them, and low contributors cannot simply free-ride without contributing something to their own model training. Personalization thus becomes a form of incentivization: each participant gets the best possible model for their data, which may require weighting the global solution towards their local optima.

- **Policy and Ethical Considerations:** Ultimately, fairness in FL for health-care has significant ethical implications. Ensuring that every hospital benefits equitably aligns with the medical ethics principle of justice. When designing federated networks, stakeholders should establish policies that explicitly address fairness. The consortium could agree on triggers that prompt model review or retraining with adjustments (such as collecting more data from underperforming sites or re-balancing the training objective). There is also an argument for transparency: participants should be kept informed about how the collaborative model is performing for each of them. Transparency can enhance trust because every partner will know that the system is doing its best to be fair and to take action to rectify any unfair outcome.

In conclusion, the evaluation of client contributions and benefits in this research provides a lens for examining fairness in federated learning for healthcare. The key takeaway is that achieving fairness is feasible but not automatic; it requires deliberate monitoring and, if needed, intervention. The intent here is to align the interests of all the individual clients with the objectives of the federation. This will drive engagement, generate larger datasets, and develop more robust, generalizable models that ultimately facilitate better care for all patients across all institutions.

7 Discussion and Future Works

7.1 Interpretation of Key Findings

This research evaluated the contributions and benefits of clients in a federated learning (FL) setting applied to healthcare data. The client’s contributions were operationalized as a decrease in global model accuracy when the client was excluded (leave-one-out evaluation). At the same time, benefit was measured as the accuracy gain a client obtained from using the federated global model compared to training in isolation. Across seven heterogeneous data scenarios using the CC-19 chest X-ray dataset, the results revealed several notable patterns.

- **Disparate Contribution vs Benefit:** Clients with small and/or noisy datasets had high benefit-to-contribution ratios, demonstrating improvement from FL, but contributed little to the global model. Clients with larger or more diverse datasets are much more likely to have made meaningful contributions to the global model. However, overall, they experienced less personal benefit than clients with small or noisy datasets.
- **Impact of Model Complexity:** Complex models such as ResNet18 and MobileNetV2 reduced fairness gaps by generalizing better across heterogeneous client data. Simpler models (TinyCNN, SimpleCNN) struggled under non-IID conditions, allowing specific clients, especially those with large datasets, to dominate the training dynamics. This suggests that architecture choice can strongly influence fairness.
- **Effect of Data Heterogeneity:** Label skew, concept shift, and feature imbalance worsened the imbalance between contributions and benefits. Specific clients with limited data or data deemed unique or critical dictated model behavior, while others benefited significantly from aggregating those models,

despite having little influence on model behavior. The severe heterogeneity from label skew led the model to be overly reliant on the marginal distributions in those clients' data, leaving clients' marginal data dependent on other contributing clients rather than substantially improving the global model's behavior.

- **Lack of Fairness Enforcement in Standard FL:** FedAvg and comparable protocols do not ensure equity: dividends and contributions are tied to data size/quality, not to equity in influence. Creating a persistent imbalance in which some clients contribute far more than they receive can slowly disincentivize participation. At a minimum, this confirms that fairness needs to be assessed and that the balance between contributions and payouts needs to be considered.

7.2 Limitations of Current Work

Despite yielding insightful observations, this study has several limitations that must be acknowledged:

- **Limited Dataset Size and Diversity:** The experiments were performed on a relatively small dataset (a few thousand grayscale chest X-ray images across three classes). The limited size and scope may not accurately reflect the complexity and variability of medical data in the real world, potentially limiting the generalizability of the results.
- **Small Number of Simulated Clients:** Only three client nodes were simulated in our federated setting. With such a small number of participants, the dynamics of contribution and fairness may differ from those in larger federations. Real healthcare consortia could involve tens of hospitals; this three-client setup limits the ability to generalize emerging fairness patterns (e.g., the distribution of free-riders vs. contributors) to larger, more diverse groups.
- **Simulated Environment (Single-Machine):** All clients were hosted on a single machine using the Flower framework; thus, the federated learning was simulated rather than truly distributed. No network latency, bandwidth restrictions, or asynchronous update components were included. The controlled

environment may therefore overlook communication delays or failures, as well as other networking factors that can impact training and fairness.

- **Public Dataset with Simplified Privacy Context:** The data used (CC-19 X-ray images) are publicly available and were treated without any explicit privacy-preserving measures in this work. This does not accurately reflect the true privacy and regulatory constraints of real-world healthcare FL, where data never leaves institutional control and legal/privacy considerations (such as HIPAA and GDPR) strongly influence system design.

7.3 Future Research Directions

Building on the insights and limitations outlined earlier, there are several promising avenues for future research to enhance federated learning in healthcare, emphasizing fairness and reliable collaboration:

- **Scaling to Larger Federations:** Future studies should simulate and experiment with larger numbers of clients (e.g., 10 or more participating hospitals or institutions) to observe fairness trends at scale. A larger federation could reveal emergent patterns (such as clusters of similar contributors or more pronounced free-rider effects) that are not apparent with only three clients.
- **Advanced and Hybrid Model Architectures:** Exploring more sophisticated neural architectures can improve overall performance and potentially fairness. Future work could include modern CNN variants with attention mechanisms (for example, transformer-based vision models or hybrid CNN-transformer backbones) that better adapt to heterogeneous data.
- **Asynchronous and Robust FL Protocols:** Future implementations ought to consider asynchronous federated learning protocols and resilience to real-world conditions. Investigating FL algorithms that tolerate client dropouts, varying update frequencies, or non-synchronized communication will make the system more applicable to realistic healthcare networks.
- **Longitudinal Studies and Client Drift:** Another important direction is to study longitudinal behavior in federated networks. Over time, client data

distributions may change; simulating FL training over extended timeframes will help assess how contribution and benefit evolve.

In summary, by addressing the limitations of this research and adopting the above directions, future work can advance FL frameworks that achieve not only high accuracy but also fairness, sustainability, and shared benefit, crucial goals for equitable healthcare collaboration.

7.4 Conclusion and Final Thoughts

7.4.1 Summary of Contributions

This research created a federated learning framework for healthcare, utilizing three simulated clients and the COVID-19 Chest X-ray dataset. To investigate the relationship between model complexity and performance in a federated context, four convolutional neural network architectures (TinyCNN, SimpleCNN, ResNet18, and MobileNetV2) were implemented and evaluated. A leave-one-out methodology was developed to quantify each client's contribution to the global model by measuring the decline in overall accuracy when the client was excluded. Similarly, each client's benefit from federated learning was assessed as the accuracy gain achieved by collaborating (federated training) compared to training in isolation. The system was examined under seven practical data heterogeneity scenarios (including label imbalance, concept drift, and data skew) by evaluating model performance and fairness metrics across multiple environments. These outcomes provided an extensive experimental baseline demonstrating how different models and datasets affect federated results.

7.4.2 Significance of the Study

The findings underscore that standard federated learning (such as basic FedAvg) does not inherently guarantee fairness in heterogeneous healthcare settings. In this research, a few specific clients received disproportionate benefits: a client with unique or extensive data might contribute substantially to the global model but

see only modest improvement in personal accuracy, suggesting a misalignment between contribution and benefit. It was found that the interplay among the model’s complexity, data distribution, and the size of the data held by clients significantly impacted this alignment. In particular, non-IID data led participants to experience reduced performance gains and a practically reduced contribution from others [16]. The practical evidence of differences in fairness justly motivates us to focus on fairness-aware strategies in FL. It is essential to ensure that clients contribute more and receive equal benefits within an ethical, sustainable collaboration. Beyond its specific findings, this study presents a reproducible template for assessing fairness in FL; the framework and the measures discussed can also serve as a guide for future researchers and practitioners to evaluate and address fairness issues in federated models.

7.4.3 Final Reflections

Overall, this research emphasizes the critical importance of fairness auditing in real-world federated learning deployments. In high-stakes domains like healthcare, it is not sufficient for a federated model to be accurate; it must also distribute its performance gains equitably among participating institutions. It is essential to align each client’s contribution with the benefit they have derived, as this influences trust and the ongoing willingness to participate in collaborative training over the long term. An unequal share of perceived performance or credit for a model can decrease motivation to participate and coalesce or divide key contributors to federated learning [16]. By raising these considerations, this research lays the stage for several possible next steps. One aspect consists of the design of incentive mechanisms and reward systems that account for contributions to motivate data sharing and fairness. Another aspect is to investigate more advanced CNN architectures or personalized FL approaches that can handle heterogeneity without sacrificing accuracy. Finally, validating these approaches in real, multi-center healthcare environments will be critical for their adoption. The findings of this study reaffirm that, as federated learning transitions from conception to deployment, fairness must be intentionally addressed to meaningfully improve healthcare performance through an equity lens, fostering sustainable, trust-based relationships.

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I confirm that I have written this thesis unaided and without using sources other than those listed and that this thesis has never been submitted to another examination authority and accepted as part of an examination achievement, neither in this form nor in a similar form. All content that was taken from a third party, either verbatim or in substance, has been acknowledged as such.

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